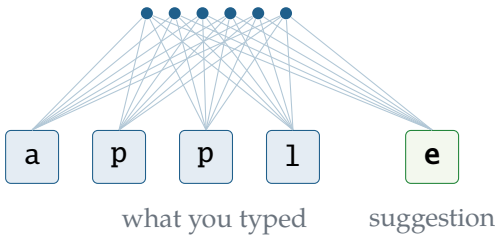
A Next-Letter Predictor

in Nex

Building a small neural network from scratch,
one bite-sized step at a time



A step-by-step tutorial

Every line of code, every equation, nothing hidden

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Chapter 1

What we are going to build

Open a terminal, type the beginning of a word, and a program suggests how to continue it, one letter at a time:

```
start of word> app
  app[r]   (23%)      others: e 19%, l 17%, o 15%, i 9%
  > l
  appl[e]   (45%)      others: i 19%, a 13%, y 11%, o 8%
  >
  apple[a]   (21%)      others: s 18%, r 11%, end 7%, n 7%
  > n
  apple[s]   (18%)      others: a 21%, r 11%, end 7%, n 7%
  >
  apples   -> looks complete (42%)      others: s 19%, t 15%, e 4%, h 4%
  >
  ==> apples
```

After `app`, the program thinks `r` is the most likely next letter (23%), with `e` and `l` close behind. We typed `l` ourselves to steer it. After `appl` it is fairly sure about `e` (45%), and we pressed Enter to accept. Then we asked for its second choice with `n`, and accepted `s`. Finally the model judged `apples` complete, and we agreed.

The thing making the suggestions is a *neural network*: a few thousand numbers, arranged in a particular way, that started out random and were nudged, millions of times, until they got good at one job. The job is to look at the last four letters of a word and give a probability for each possible next letter.

This is, in miniature, exactly what a large language model does. A chatbot looks at the text so far and gives a probability for each possible next piece of text. The pieces are bigger (whole words or parts of words, not letters), the network is far bigger (billions of numbers, not thousands), and it looks much further back than four letters. But the recipe is the same, and every ingredient of the recipe is in this small project:

- turning symbols into vectors of numbers (*embeddings*);
- layers of artificial neurons;
- turning scores into probabilities (*softmax*);
- measuring how wrong a prediction is (*cross-entropy loss*);
- working out how to change every number to be a little less wrong (*backpropagation*);
- changing them, over and over (*gradient descent*).

1.1 How this tutorial works

We will build the whole project together, from an empty folder, in small steps. Each step adds a few lines of code to one of the project's files, and explains what the code does and why. When there is math behind the code, we work through the math first, in simple steps, and then turn it into code.

The code comes in boxes like this one. The title of the box says which file the code belongs to, and where in the file it goes:

New file `example.nex`

Start a new file with this content.

`example.nex` – add at the end

Add this code at the end of the file.

`example.nex` – add before the line ...

Add this code at a particular place in the file, which the title tells you.

`example.nex` – continued

Add this code right after the code from the previous box.

Green boxes are *try-it* scripts: small throwaway programs that use what we just built, so you can see it working. Their output follows in a dark box:

Try it `try_example.nex`

A small program to run now. It is not part of the project.

```
$ nex try_example.nex
what it prints
```

Every program and every output in this tutorial was produced by running the code exactly as printed here, so if your output differs, compare your code with the boxes. (Training on a different word list gives different numbers, of course.)

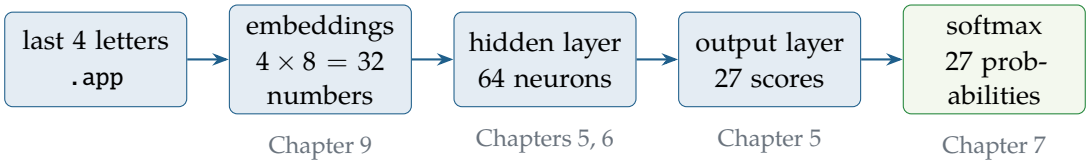
1.2 What you need

- **Nex**, installed so that the `nex` command works in a terminal. You should know the basics of the language: `let`, functions and `result`, classes, loops, and contracts (`require`, `ensure`, `invariant`). We explain the less common features as we meet them.
- **A word list**: a text file with one word per line. Linux and macOS have one at `/usr/share/dict/words`. Any other list works too.
- **Math**: school algebra. We will use vectors, matrices, the exponential and logarithm, and derivatives, and we explain each one when it first comes up. You don't need to have seen any of it in a long time.

No machine-learning library is used. Every multiplication, every derivative and even the random number generator is written out in Nex, in about 1,400 lines including comments and checks.

1.3 The plan

Here is the whole network, which the following chapters build one piece at a time:



2–3. Set up the project, and decide how to represent letters.
4–7. The math toolkit: vectors, matrices, a squashing function, and softmax.
8–10. Random numbers, the model itself, and running it forward.
11–15. Training: the data, the loss, gradient descent, backpropagation, and checking that the gradients are right.
16–18. Saving models, the training program, and the program you play with.

By the end you will have these files, each of which you will have written yourself:

File	What it holds
`helpers.nex`	Printing numbers neatly, and finding files
`vocabulary.nex`	The 27 symbols the model knows
`vector_math.nex`	Dot products, layers, tanh, softmax
`rng.nex`	A random number generator
`next_letter_model.nex`	The neural network: forward pass, backpropagation, update
`training_data.nex`	Training examples, the loss, and evaluation
`checks.nex`	Automatic checks of everything, including the gradients
`train.nex`	The training program
`play.nex`	The word-completion program

Chapter 2

Setting up

Make a folder for the project and go into it. Every file in this tutorial goes in this one folder.

```
mkdir nextletter
cd nextletter
```

2.1 A first file: printing numbers

We will print a lot of numbers along the way: probabilities, losses, weights. Nex prints a `Real` with all its digits, like `0.6652409557748218`, which is hard to read in a table. So the very first file holds a few helpers for printing numbers neatly.

A Nex program uses another file with `intern`. `intern Helpers` finds `helpers.nex` in the same folder and loads everything in it. The file does not need a class called `Helpers`: ours holds only plain functions.

The first helper writes a number with a fixed number of decimal places. We multiply by 10^{places}, round to a whole number, and then split that whole number back into the part before the point and the part after it. For `fixed(2.5, 3)`: $2.5 \times 1000 = 2500$, so the whole part is $2500/1000 = 2$, the fraction is $2500 \bmod 1000 = 500$, and the answer is `"2.500"`. The `from`… `until` loop puts back leading zeros the fraction needs: with 3 places, a fraction of `5` must be written `005`.

New file `helpers.nex`

```
-- helpers.nex: printing numbers neatly, and finding the user's files.

-- r with exactly `places` digits after the decimal point: fixed(2.5, 3)
-- is "2.500".
function fixed(r: Real, places: Integer): String
  require
    non_negative: places >= 0
  do
    let scale: Integer := 10 ^ places
    let scaled: Integer := (r.abs * scale.to_real).round
    let whole: String := (scaled / scale).to_string
    let frac: String := (scaled % scale).to_string
    from
    until
      frac.length >= places
    do
      frac := "0" + frac
    end
    result := when r < 0.0 and scaled > 0 then "-" else "" end + whole
    if places > 0 then
      result := result + "." + frac
    end
  end
```

Notice `when ... then ... else ... end`: it is an `if` that gives a value. We work on the absolute value `r.abs` and put the minus sign back at the end, but only if the rounded number isn't zero, so that `-0.0001` prints as `0.000` rather than `-0.000`.

Three more small helpers: a probability as a percentage, and padding text to a fixed width, so that columns line up.

`helpers.nex` — add at the end

```
-- A probability as a whole percentage: percent(0.257) is "26%".
function percent(p: Real): String do
  result := (p * 100.0).round.to_string + "%"
end

-- s padded with spaces on the left to at least `width` characters.
function pad_left(s: String, width: Integer): String do
  result := s
  from
  until
    result.length >= width
  do
    result := " " + result
  end
end

-- s padded with spaces on the right to at least `width` characters.
function pad_right(s: String, width: Integer): String do
  result := s
  from
  until
    result.length >= width
  do
    result := result + " "
  end
end
```

And one that prints a whole list of numbers:

`helpers.nex` — add at the end

```
-- A list of numbers as text, each with `places` decimals:
-- vector_text([0.5, 0.25], 2) is "[0.50, 0.25]".
function vector_text(v: Array[Real], places: Integer): String do
  result := "["
  across v as x do
    if result /= "[" then
      result := result + ", "
    end
    result := result + fixed(x, places)
  end
  result := result + "]"
end
```

2.2 Running a try-it script

Let's check that it works. Make this file and run it:

Try it `try_helpers.nex`

```
intern Helpers

let con: Console := create Console
con.print_line(fixed(2.5, 3))
con.print_line(fixed(-3.14159, 2))
con.print_line(percent(0.257))
con.print_line("[" + pad_left("42", 6) + "]")
con.print_line(vector_text([0.5, 0.25, 1.0 / 3.0], 3))
```

```
$ nex try_helpers.nex

2.500
-3.14
26%
[    42]
[0.500, 0.250, 0.333]
```

We print with `con.print_line` from a `Console` object, because Nex's `print` shows strings with their quotes (`"2.500"`), which is handy for debugging but not for neat output.

Try-it scripts are not part of the project. Keep them around to experiment with, or delete them when you are done.

Chapter 3

Symbols

A neural network only works with numbers. Before anything else, we decide how letters become numbers.

Our model will predict one of 27 *symbols*: the 26 letters `a` … `z`, and one extra symbol, a full stop `.`, which does two jobs.

- **It marks the end of a word.** To know when a word is finished, the model must be able to predict "the word ends here". After `apple`, the right answer is not a letter but `.`.
- **It pads the start of a word.** The model always looks at the last four symbols. At the start of a word there aren't four yet, so we fill the gap with full stops: before the first letter of `apple`, the model sees `....`, and the right answer is `a`. After `ap` it sees `..ap`.

Each symbol gets a number, its *id*: `.` is 0, `a` is 1, and so on up to `z`, which is 26. A Nex class can hold constants, written `NAME = value`, which we read through the class name: `Vocabulary.SIZE`.

New file `vocabulary.nex`

```
-- vocabulary.nex: the 27 symbols the model reads and predicts.
--
-- "." does two jobs:
--   * padding at the start, so even the first letter has a full context
--     ("...." -> "a" means "a word can start with a");
--   * the end-of-word marker, so the model can learn when to stop
--     ("appl" -> "e", then "pple" -> "." means "apple is complete").

class Vocabulary
  feature
    END = "."
    SYMBOLS = ".abcdefghijklmnopqrstuvwxyz"
    SIZE = 27
end
```

The id of a symbol is simply its position in `SYMBOLS`, and the symbol with a given id is the one-character piece of `SYMBOLS` at that position:

`vocabulary.nex` – add at the end

```
-- The id of a symbol: "." is 0, "a" is 1, ..., "z" is 26.
function symbol_id(ch: String): Integer
  require
    one_symbol: ch.length = 1 and Vocabulary.SYMBOLS.contains(ch)
  do
    result := Vocabulary.SYMBOLS.index_of(ch)
  ensure
    in_range: result >= 0 and result < Vocabulary.SIZE
  end

-- The symbol with a given id.
```

```
function symbol_at(id: Integer): String
  require
    in_range: id >= 0 and id < Vocabulary.SIZE
  do
    result := Vocabulary.SYMBOLS.substring(id, id + 1)
  ensure
    one_symbol: result.length = 1
  end
```

The contracts say exactly what these functions accept and promise. `symbol_id` only accepts a single character that is one of our symbols, and promises an id from 0 to 26. If some other part of the program ever passes it a capital letter, the program stops right there, with a message that names the broken promise, instead of carrying on with a nonsense id and failing somewhere far away.

The word list we will train on contains words such as `Paris` and `it's`, which use characters outside our 27 symbols. We'll skip those, so we need a test for "plain lowercase word":

`vocabulary.nex` – add at the end

```
-- True when every character of s is one of a..z.
function is_lowercase_word(s: String): Boolean do
  result := s.length > 0
  across s as ch do
    if ch < #a or ch > #z then
      result := false
    end
  end
end
```

`across s as ch` walks through the characters of a string. Each `ch` is a `Char`, and `#a` is how Nex writes the character `a`. Characters compare by their code, and `a` … `z` are consecutive codes, so the test is a simple range check.

Try it `try_vocabulary.nex`

```
intern Vocabulary

let con: Console := create Console
con.print_line("symbols: " + Vocabulary.SYMBOLS)
con.print_line("id of a: " + symbol_id("a").to_string)
con.print_line("id of p: " + symbol_id("p").to_string)
con.print_line("symbol 16: " + symbol_at(16))
con.print_line("apple: " + is_lowercase_word("apple").to_string)
con.print_line("Paris: " + is_lowercase_word("Paris").to_string)
do
  con.print_line(symbol_id("A").to_string)
rescue
  con.print_line("symbol_id(\"A\") refused: " + exception.to_string)
end
```

```
$ nex try_vocabulary.nex

symbols: .abcdefghijklmnopqrstuvwxyz
```

```
id of a: 1
id of p: 16
symbol 16: p
apple: true
Paris: false
symbol_id("A") refused: Precondition violation: one_symbol
```

The last lines show a contract at work: `symbol_id("A")` breaks the precondition `one_symbol`, and the `rescue` block catches the failure so that we can print it.

Chapter 4

Vectors

A neural network is mostly multiplication and addition, organized with vectors and matrices. This chapter and the next build the small amount of linear algebra we need.

4.1 The math

A *vector* is a list of numbers. We write vectors in bold: a vector $\mathbf{u}$ with n numbers, or *components*, is

$$\mathbf{u} = (u_0, u_1, \ldots, u_{n-1}).$$

We count from 0, as Nex arrays do. "$\mathbf{u}$ is in $\mathbb{R}^n$" is the short way to say "$\mathbf{u}$ is a list of n real numbers".

The single most important operation is the *dot product* of two vectors of the same length: multiply them component by component, and add up the results.

Dot product

$$\mathbf{u} \cdot \mathbf{v} \;=\; u_0 v_0 + u_1 v_1 + \cdots + u_{n-1} v_{n-1} \;=\; \sum_{i=0}^{n-1} u_i v_i$$

For example, $(1,2,3) \cdot (4,5,6) = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32$.

The $\sum$ sign (capital sigma) means "add up": $\sum_{i=0}^{n-1} u_i v_i$ reads "the sum of $u_i v_i$ for i from 0 to $n-1$".

Here is a way to think about the dot product that we will use all the time. Suppose $\mathbf{v}$ holds some *inputs*, and $\mathbf{u}$ holds a *weight* for each input. Then $\mathbf{u} \cdot \mathbf{v}$ is a *weighted sum* of the inputs: an input with a large positive weight pushes the total up a lot, an input with a negative weight pulls it down, and an input with weight zero is ignored. A shopping receipt is a dot product: quantities times prices, added up.

4.2 The code

In Nex, a vector is an `Array[Real]`. Start the math file with a class `Vector_Math` that holds one constant, which we will need in Chapter 7.

New file `vector_math.nex`

```
-- vector_math.nex: the few pieces of linear algebra the network needs.
--
-- A vector is an Array[Real]. A matrix is an Array of rows, where each row
-- is a vector. No library does the math for us: every multiply and add is
-- written out below.

class Vector_Math
  feature
```

```
    -- How far a sum of probabilities may drift from 1.0 through rounding.
    TOLERANCE = 1.0e-9
end
```

Two simple functions first: a vector of zeros, and the sum of a vector's components.

vector_math.nex — add at the end

```
-- A vector of n zeros.
function zeros(n: Integer): Array[Real]
  require
    non_negative: n >= 0
  do
    result := create Array.filled(n, 0.0)
  ensure
    right_length: result.length = n
  end

-- The sum of the elements.
function sum(v: Array[Real]): Real do
  across v as x do
    result := result + x
  end
end
```

`sum` relies on a Nex rule: `result` starts at the zero value of its type, which is `0.0` for `Real`, so there is no need to set it before the loop.

The dot product is the formula above, as a loop. Its precondition says the two vectors must have the same length; otherwise the dot product isn't defined.

vector_math.nex — add at the end

```
-- The dot product: u[0]*v[0] + u[1]*v[1] + ...
function dot(u, v: Array[Real]): Real
  require
    same_length: u.length = v.length
  do
    from
      let i: Integer := 0
    until
      i = u.length
    do
      result := result + u.get(i) * v.get(i)
      i := i + 1
    end
  end
```

Try it try_vectors.nex

```
intern Vector_Math

let con: Console := create Console
let u: Array[Real] := [1.0, 2.0, 3.0]
let v: Array[Real] := [4.0, 5.0, 6.0]
```

```
con.print_line("zeros(3)  = " + zeros(3).to_string)
con.print_line("sum(u)    = " + sum(u).to_string)
con.print_line("dot(u, v) = " + dot(u, v).to_string)
do
  con.print_line(dot([1.0, 2.0], [1.0]).to_string)
rescue
  con.print_line("dot of different lengths: " + exception.to_string)
end
```

```
$ nex try_vectors.nex

zeros(3)  = [0.0, 0.0, 0.0]
sum(u)    = 6.0
dot(u, v) = 32.0
dot of different lengths: Precondition violation: same_length
```

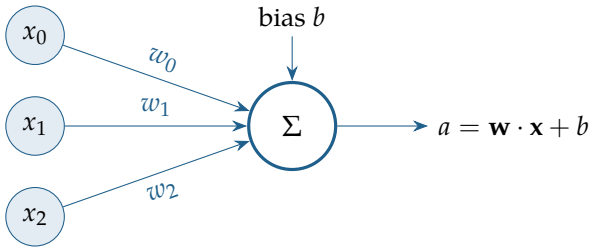
Chapter 5

Matrices and layers

5.1 One neuron

An artificial *neuron* takes some inputs $x_0, \ldots, x_{n-1}$, and has a weight w_j for each input and one extra number b, its *bias*. It computes the weighted sum of its inputs and adds the bias:

$$a = w_0 x_0 + w_1 x_1 + \cdots + w_{n-1} x_{n-1} + b = \mathbf{w} \cdot \mathbf{x} + b.$$



The weights decide which inputs the neuron cares about, and the bias shifts its output up or down regardless of the inputs. Training a network means finding good weights and biases.

5.2 A layer of neurons is a matrix

A *layer* is several neurons that all read the same inputs, each with its own weights and bias. If we write each neuron's weights as one row, we get a table of numbers with one row per neuron and one column per input. Such a table is a *matrix*. A matrix with m rows and n columns is an "$m \times n$ matrix". The number in row i, column j of a matrix W is written W_{ij}.

Running the whole layer means taking the dot product of every row with the input, and adding each row's bias. That is the *matrix–vector product* $W\mathbf{x}$, plus a bias vector $\mathbf{b}$:

A layer: $\mathbf{a} = W\mathbf{x} + \mathbf{b}$

For an $m \times n$ matrix W, an input $\mathbf{x} \in \mathbb{R}^n$ and biases $\mathbf{b} \in \mathbb{R}^m$, the output $\mathbf{a} \in \mathbb{R}^m$ has one component per row:

$$a_i = \sum_{j=0}^{n-1} W_{ij}\, x_j + b_i = (\text{row } i \text{ of } W) \cdot \mathbf{x} + b_i.$$

For example,

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 0.5 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 0 \cdot 3 + 0.5 \\ 1 \cdot 2 + 1 \cdot 3 - 1 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 4 \end{pmatrix}.$$

A function of the form $W\mathbf{x} + \mathbf{b}$ is called *affine*. Our network will have two layers like this: one with 64 neurons and 32 inputs (a 64×32 matrix), and one with 27 neurons, one per symbol, and 64 inputs (a 27×64 matrix).

5.3 The code

In Nex, a matrix is an array of rows: `Array[Array[Real]]`. First, a matrix of zeros. It builds each row with its own call to `zeros`, so that every row is a separate array. (If all rows were the same array, changing one would change them all.)

vector_math.nex — add at the end

```
-- A rows x cols matrix of zeros. Each row is its own array, so changing
-- one row never changes another.
function zero_matrix(rows, cols: Integer): Array[Array[Real]]
  require
    non_negative: rows >= 0 and cols >= 0
  do
    repeat rows do
      result.add(zeros(cols))
    end
  ensure
    right_rows: result.length = rows
  end
```

Then the layer itself. It is a dot product per row, exactly as in the box above:

vector_math.nex — add at the end

```
-- W·x + b: one dot product per row of W, plus that row's bias.
-- Each row is one neuron; all the rows together are a layer.
function affine(w: Array[Array[Real]], x, b: Array[Real]): Array[Real]
  require
    one_bias_per_row: w.length = b.length
  do
    from
      let i: Integer := 0
    until
      i = w.length
    do
      result.add(dot(w.get(i), x) + b.get(i))
      i := i + 1
    end
  ensure
    one_output_per_row: result.length = w.length
  end
```

One more operation, which looks modest but will do most of the work in training: add a multiple of one vector to another, in place, $\mathbf{t} \leftarrow \mathbf{t} + s\,\mathbf{v}$. (The arrow means "becomes".)

vector_math.nex — add at the end

```
-- target := target + s * v, changing target in place.
-- Backpropagation and gradient descent are built almost entirely from this.
function add_scaled(target: Array[Real], s: Real, v: Array[Real])
  require
    same_length: target.length = v.length
  do
    from
      let i: Integer := 0
```

```
    until
      i = target.length
    do
      target.set(i, target.get(i) + s * v.get(i))
      i := i + 1
    end
  end
```

Try it `try_layer.nex`

```
intern Vector_Math
intern Helpers

let con: Console := create Console
-- The example from the box: two neurons with two inputs each.
let w: Array[Array[Real]] := [[1.0, 0.0], [1.0, 1.0]]
let x: Array[Real] := [2.0, 3.0]
let b: Array[Real] := [0.5, -1.0]
con.print_line("W x + b = " + vector_text(affine(w, x, b), 1))

let t: Array[Real] := [1.0, 1.0]
add_scaled(t, 0.5, [2.0, 4.0])
con.print_line("[1, 1] + 0.5 * [2, 4] = " + vector_text(t, 1))
con.print_line("a 2 x 3 zero matrix: " + zero_matrix(2, 3).to_string)
```

```
$ nex try_layer.nex

W x + b = [2.5, 4.0]
[1, 1] + 0.5 * [2, 4] = [2.0, 3.0]
a 2 x 3 zero matrix: [[0.0, 0.0, 0.0], [0.0, 0.0, 0.0]]
```

Chapter 6

Bending the line: tanh

6.1 Why layers alone are not enough

If we put two affine layers one after the other, something disappointing happens. Feed $\mathbf{x}$ to the first layer, and its output to the second:

$$W_2(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 \;=\; (W_2W_1)\,\mathbf{x} + (W_2\mathbf{b}_1 + \mathbf{b}_2).$$

The result is again of the form (some matrix)$\cdot\mathbf{x}$ + (some vector): two affine layers are no more powerful than one. However many we stack, we only ever get an affine function, and affine functions can only describe straight-line ("linear") relationships. Predicting spelling needs much more than that.

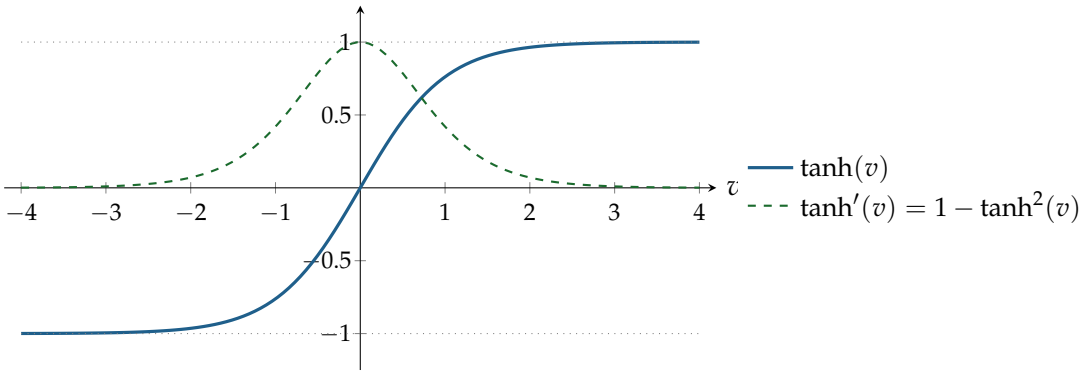
The fix is to pass each neuron's output through a *nonlinear* function before the next layer. This function is called the *activation function*, and it is what lets a network describe curved, complicated relationships. We use the *hyperbolic tangent*, tanh.

6.2 The math of tanh

The hyperbolic tangent

$$\tanh(v) = \frac{e^{v} - e^{-v}}{e^{v} + e^{-v}}$$

Here $e \approx 2.718$ is Euler's number, and e^{v} is the *exponential* function: it is always positive, it is 1 at $v = 0$, it grows very fast for positive v ($e^{10} \approx 22026$), and it shrinks toward 0 for negative v.



The solid curve is tanh. It is close to a straight line near zero ($\tanh(v) \approx v$ for small v), and it flattens out toward $+1$ for large positive v and toward -1 for large negative v. So it *squashes* any number into the range $(-1, 1)$. A neuron followed by tanh behaves like a soft switch: strongly negative input means "off" (-1), strongly positive input means "on" ($+1$), and in between it responds smoothly.

A form that is safe to compute

The formula above has a practical problem. For $v = 1000$, e^{1000} is far too big for a computer's floating-point numbers: it becomes `Infinity`, and ∞/∞ is not a number at all. Let's rewrite the formula. Multiply the top and the bottom by e^{v}:

$$\tanh(v) = \frac{e^{2v} - 1}{e^{2v} + 1} = \frac{(e^{2v} + 1) - 2}{e^{2v} + 1} = 1 - \frac{2}{e^{2v} + 1}.$$

The last form is safe. For large v, e^{2v} becomes `Infinity`, $2/\infty = 0$, and the answer is exactly 1, which is correct. For very negative v, e^{2v} becomes 0, and the answer is $1 - 2 = -1$, also correct.

The derivative

Training will need to know how fast tanh changes, which is its *derivative*. (Chapter 12 reviews derivatives; for now, the derivative of a function at a point is the slope of its graph there.) The derivative of tanh has a remarkably simple form:

Derivative of tanh

$$\tanh'(v) = 1 - \tanh^2(v)$$

To see why, write $\tanh(v) = s/c$ with $s = e^{v} - e^{-v}$ and $c = e^{v} + e^{-v}$. Since the derivative of e^{v} is e^{v} and the derivative of e^{-v} is $-e^{-v}$, we get $s' = c$ and $c' = s$. The quotient rule $(s/c)' = (s'c - sc')/c^2$ gives

$$\tanh'(v) = \frac{c \cdot c - s \cdot s}{c^2} = 1 - \frac{s^2}{c^2} = 1 - \tanh^2(v).$$

The dashed curve in the plot is this derivative. It is 1 at $v = 0$, where tanh is steepest, and nearly 0 far from zero, where tanh is flat. The nice thing about this form is that once we have computed $h = \tanh(v)$, the derivative costs one multiplication: $1 - h^2$.

6.3 The code

`Real` in Nex has `exp`, but no `tanh`, so we write it ourselves, using the safe form:

`vector_math.nex` – add at the end

```
-- The hyperbolic tangent, squashing any real number into (-1, 1).
-- Real has no tanh method, so we build it from exp:
--     tanh(v) = (e^2v - 1) / (e^2v + 1) = 1 - 2 / (e^2v + 1)
-- The second form is safe for large v: e^2v overflows to Infinity,
-- 2 / Infinity is 0, and the answer is exactly 1.
function tanh(v: Real): Real
  do
    result := 1.0 - 2.0 / ((2.0 * v).exp + 1.0)
  ensure
    bounded: result >= -1.0 and result <= 1.0
  end
```

Try it `try_tanh.nex`

```
intern Vector_Math
intern Helpers

let con: Console := create Console
con.print_line("     v    tanh(v)   1 - tanh(v)^2")
across [-3.0, -1.0, -0.1, 0.0, 0.1, 1.0, 3.0] as v do
  let h: Real := tanh(v)
  con.print_line(pad_left(fixed(v, 1), 6) + pad_left(fixed(h, 4), 11)
    + pad_left(fixed(1.0 - h * h, 4), 16))
end
con.print_line("tanh(1000)  = " + tanh(1000.0).to_string)
con.print_line("tanh(-1000) = " + tanh(-1000.0).to_string)
```

```
$ nex try_tanh.nex

     v    tanh(v)   1 - tanh(v)^2
  -3.0    -0.9951          0.0099
  -1.0    -0.7616          0.4200
  -0.1    -0.0997          0.9901
   0.0     0.0000          1.0000
   0.1     0.0997          0.9901
   1.0     0.7616          0.4200
   3.0     0.9951          0.0099
tanh(1000)  = 1.0
tanh(-1000) = -1.0
```

Near zero, $\tanh(0.1) \approx 0.0997$, almost exactly 0.1: there, tanh is nearly a straight line. At ± 3, the derivative has dropped to about 0.01: the curve is almost flat.

Chapter 7

From scores to probabilities: softmax

The last layer of our network has 27 neurons, one per symbol. Their outputs are 27 arbitrary numbers, called *scores* or *logits*: a high score means "this symbol is likely". But what we want is probabilities: 27 numbers between 0 and 1 that add up to 1.

7.1 The math

The *softmax* function does this conversion. It makes every score positive by taking its exponential, and then divides by the total, so the results add up to 1:

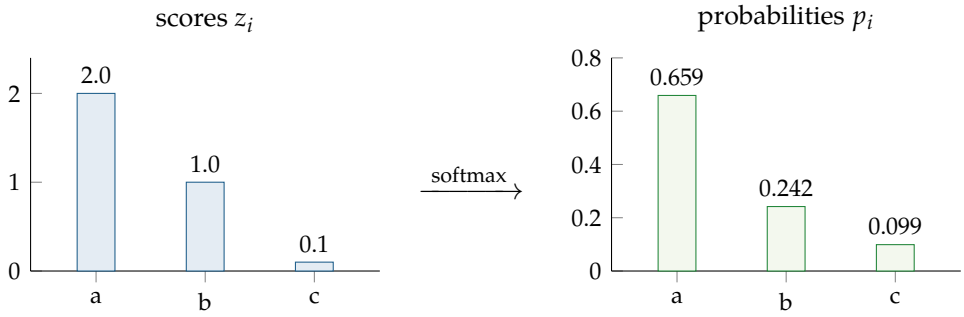
Softmax

For scores $\mathbf{z} = (z_0, \ldots, z_{K-1})$,

$$p_i = \text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=0}^{K-1} e^{z_j}}.$$

Every p_i is positive, and $\sum_i p_i = 1$.

For example, take scores $(2.0, 1.0, 0.1)$. The exponentials are $e^{2.0} \approx 7.389$, $e^{1.0} \approx 2.718$ and $e^{0.1} \approx 1.105$, which add up to 11.212. Dividing each by the total gives the probabilities $(0.659, 0.242, 0.099)$.



Notice that the exponential exaggerates differences: the scores 2.0 and 1.0 differ by 1, and their probabilities differ by a factor of $e^1 \approx 2.7$. In general, adding 1 to a score multiplies that symbol's (unnormalized) weight by e.

Shifting the scores changes nothing

If we add the same number c to every score, the probabilities stay exactly the same:

$$\frac{e^{z_i+c}}{\sum_j e^{z_j+c}} = \frac{e^c\, e^{z_i}}{e^c \sum_j e^{z_j}} = \frac{e^{z_i}}{\sum_j e^{z_j}},$$

because e^c appears on the top and on every term of the bottom, and cancels. This matters for the same reason as with tanh: e^z overflows to infinity once z is bigger than about 709. So before

taking exponentials, we subtract the largest score, $m = \max_j z_j$, from every score. Then the largest exponent is $e^0 = 1$, nothing can overflow, and the answer is unchanged.

7.2 The code

vector_math.nex — add at the end

```
-- Turn raw scores ("logits") into probabilities that sum to 1.
-- Subtracting the largest score first changes nothing in the answer
-- (it cancels between top and bottom), but it keeps exp from
-- overflowing when a score is large.
function softmax(z: Array[Real]): Array[Real]
  require
    not_empty: z.length > 0
  do
    let m: Real := z.get(0)
    across z as v do
      m := m.max(v)
    end
    let exps: Array[Real] := []
    across z as v do
      exps.add((v - m).exp)
    end
    let total: Real := sum(exps)
    across exps as e do
      result.add(e / total)
    end
  ensure
    same_length: result.length = z.length
    sums_to_one: (sum(result) - 1.0).abs < Vector_Math.TOLERANCE
  end
```

The postcondition `sums_to_one` is where the `TOLERANCE` constant comes in. Floating-point arithmetic rounds a little at every step, so the sum may come out as `0.9999999999999999` instead of exactly 1. The contract allows a tiny difference, and nothing more. Every time the network runs, this contract checks that our probabilities really are probabilities.

We will also want the position of the largest number in a vector: the network's top choice.

vector_math.nex — add at the end

```
-- The index of the largest element (the first one, if there is a tie).
function argmax(v: Array[Real]): Integer
  require
    not_empty: v.length > 0
  do
    from
      let i: Integer := 1
    until
      i = v.length
    do
      if v.get(i) > v.get(result) then
        result := i
      end
      i := i + 1
```

```
    end
  ensure
    in_range: result >= 0 and result < v.length
  end
```

Try it `try_softmax.nex`

```
intern Vector_Math
intern Helpers

let con: Console := create Console
let p: Array[Real] := softmax([2.0, 1.0, 0.1])
con.print_line("softmax([2.0, 1.0, 0.1]) = " + vector_text(p, 3))
con.print_line("sum = " + sum(p).to_string)
let shifted: Array[Real] := softmax([1002.0, 1001.0, 1000.1])
con.print_line("scores shifted by 1000  = " + vector_text(shifted, 3))
con.print_line("equal scores            = " + vector_text(softmax([5.0, 5.0, 5.0, 5.0]), 3))
con.print_line("argmax = " + argmax(p).to_string)
```

```
$ nex try_softmax.nex

softmax([2.0, 1.0, 0.1]) = [0.659, 0.242, 0.099]
sum = 1.0
scores shifted by 1000  = [0.659, 0.242, 0.099]
equal scores            = [0.250, 0.250, 0.250, 0.250]
argmax = 0
```

Scores of 1000 and more would overflow without the shift; with it, we get the same probabilities as before. And equal scores give equal probabilities, as they should.

Chapter 8

Randomness

A new network starts with random weights. (If all weights started equal, all the neurons in a layer would compute the same thing, receive the same updates, and stay identical forever. Randomness breaks the tie.) Training also picks examples at random. We want this randomness to be *repeatable*: two runs with the same *seed* should produce exactly the same "random" numbers, so that a result can be reproduced and a bug can be chased. So we write our own small random number generator.

8.1 Uniform numbers: Park–Miller

The idea of a *pseudo-random* generator is to keep a hidden number, the *state*, and to update it with a fixed formula every time we want a new number. The formula is chosen so that the sequence of states looks random. One of the simplest good formulas is the Park–Miller "minimal standard" generator:

Park–Miller generator

$$\text{state} \leftarrow (\text{state} \times 48271) \bmod M, \qquad M = 2^{31} - 1 = 2147483647.$$

"a mod M" is the remainder after dividing a by M, written `a % M` in Nex. Because M is a prime number, and 48271 is carefully chosen, the state runs through every number from 1 to $M - 1$, in a scrambled order, before it repeats. It must never be 0, since 0 would stay 0 forever.

There is one more thing to check. Nex's `Integer` has 64 bits, and arithmetic that goes past 2^{63} is an error, not a silent wrap-around. The state is always less than $M < 2^{31}$, and $48271 < 2^{16}$, so the product is less than 2^{47}. It always fits.

To get a real number between 0 and 1, divide the state by M. Since the state is between 1 and $M - 1$, the result is strictly between 0 and 1, never exactly 0 or 1. (We will take its logarithm in a moment, and $\ln 0$ is $-\infty$, so this matters.)

New file `rng.nex`

```
-- rng.nex: a small random number generator that can be seeded.
--
-- Neural networks start from random weights and learn from randomly
-- chosen examples. With a seed, the "random" numbers are the same on every
-- run, so a training run can be repeated exactly.
--
-- This is the Park-Miller "minimal standard" generator:
--     state := state * 48271 mod (2^31 - 1)
-- 2^31 - 1 is prime, and every state stays below it, so state * 48271 is
-- less than 2^47 and never overflows a 64-bit Integer.

class Rng
  create
    make(seed: Integer) do
```

```
      -- Map any seed into the valid range 1 .. MODULUS - 1 (0 would stick).
      state := 1 + seed.abs % (MODULUS - 1)
    end
  feature
    MODULUS = 2147483647
    MULTIPLIER = 48271

    state: Integer

    -- A real number spread evenly over the open interval (0, 1).
    uniform(): Real
      do
        state := (state * MULTIPLIER) % MODULUS
        result := state.to_real / MODULUS.to_real
      ensure
        open_interval: result > 0.0 and result < 1.0
      end

  invariant
    state_in_range: state >= 1 and state < MODULUS
end
```

The class invariant `state_in_range` is checked after the object is created and around every call to it, so a state of 0 could never sneak in.

8.2 Whole numbers

To pick a random example out of n, we need a whole number from 0 to $n - 1$. Multiply a uniform number $u \in (0, 1)$ by n and round down: $\lfloor u\, n \rfloor$. Since $0 < u\, n < n$, the result is one of $0, 1, \ldots, n - 1$, each about equally likely.

`rng.nex` – add before the invariant

```
    -- A whole number in 0 .. n - 1, each equally likely.
    below(n: Integer): Integer
      require
        positive: n > 0
      do
        result := (uniform() * n.to_real).floor
      ensure
        in_range: result >= 0 and result < n
      end
```

8.3 Bell-curve numbers: Box–Muller

Starting weights are usually drawn from the *normal distribution*, the familiar bell curve: most values near 0, fewer and fewer further out. A *standard* normal number has mean (average) 0 and standard deviation 1, which means that about 68% of the values are between -1 and 1, and about 95% are between -2 and 2.

The Box–Muller transform makes two standard normal numbers out of two uniform numbers $u_1, u_2 \in (0, 1)$:

Box–Muller transform

$$r = \sqrt{-2 \ln u_1}, \qquad \theta = 2\pi u_2, \qquad z_1 = r \cos\theta, \qquad z_2 = r \sin\theta.$$

Why does this work? Picture a point (z_1, z_2) whose two coordinates are independent standard normal numbers: a cloud of points, densest at the origin. The cloud is perfectly round, so the direction θ of a point is equally likely to be any angle: that is $2\pi u_2$. Its distance r from the origin is not uniform; a calculation in polar coordinates shows that r^2 follows the *exponential* distribution with mean 2, and $-2 \ln u_1$ is exactly how to make such a number from a uniform one. Then $r \cos\theta$ and $r \sin\theta$ turn the polar coordinates back into z_1 and z_2.

The transform gives two numbers at a time. We return the first and keep the second, the *spare*, for the next call:

`rng.nex` – add before the invariant

```
    -- Box-Muller (below) makes normal numbers in pairs; this holds the
    -- second one until it is asked for.
    has_spare: Boolean
    spare: Real

    -- A number from the standard normal ("bell curve") distribution, with
    -- mean 0 and standard deviation 1, by the Box-Muller transform: two
    -- uniform numbers u1, u2 give two independent normal numbers
    --     sqrt(-2 ln u1) * cos(2 pi u2)   and   sqrt(-2 ln u1) * sin(2 pi u2)
    gauss(): Real do
      if has_spare then
        has_spare := false
        result := spare
      else
        let radius: Real := (-2.0 * uniform().log).sqrt
        let angle: Real := 2.0 * Real.PI * uniform()
        spare := radius * angle.sin
        has_spare := true
        result := radius * angle.cos
      end
    end
```

8.4 Shuffling

Finally, we need to shuffle a list: to split words fairly into a training set and a test set. The *Fisher–Yates* shuffle walks from the last position to the first, and swaps the element at each position i with a randomly chosen element at a position j from 0 to i (possibly i itself). The last position gets each of the n elements with equal chance, the one before it gets each of the remaining $n - 1$, and so on, so each of the $n!$ possible orders is equally likely.

The function is *generic*: `[T]` means it works on an array of anything, words or training examples. (Nex allows generic functions but not generic methods, so it is a function that takes the generator as an argument.)

rng.nex — add at the end

```
-- Shuffle an array in place (Fisher-Yates): walk from the end, swapping
-- each element with a randomly chosen one at or before it.
function shuffle[T](rng: Rng, items: Array[T]) do
  from
    let i: Integer := items.length - 1
  until
    i <= 0
  do
    let j: Integer := rng.below(i + 1)
    let tmp: T := items.get(i)
    items.set(i, items.get(j))
    items.set(j, tmp)
    i := i - 1
  end
end
```

Try it try_rng.nex

```
intern Rng
intern Helpers

let con: Console := create Console
let rng: Rng := create Rng.make(42)
con.print_line("uniform: " + fixed(rng.uniform(), 4) + " " + fixed(rng.uniform(), 4))
con.print_line("below(10): " + rng.below(10).to_string + " " + rng.below(10).to_string)

-- Draw many normal numbers and measure their mean and spread.
let n: Integer := 100000
let total: Real := 0.0
let squares: Real := 0.0
let within_one: Integer := 0
repeat n do
  let z: Real := rng.gauss()
  total := total + z
  squares := squares + z * z
  if z.abs < 1.0 then
    within_one := within_one + 1
  end
end
let mean: Real := total / n.to_real
con.print_line("mean " + fixed(mean, 3) + ", standard deviation "
  + fixed((squares / n.to_real - mean * mean).sqrt, 3)
  + ", within 1 of the mean: " + percent(within_one.to_real / n.to_real))

let words: Array[String] := ["one", "two", "three", "four", "five"]
shuffle(rng, words)
con.print_line("shuffled: " + words.to_string)
let again: Rng := create Rng.make(42)
con.print_line("same seed again: " + fixed(again.uniform(), 4))
```

```
$ nex try_rng.nex

uniform: 0.0010 0.6564
below(10): 8 3
```

```
mean 0.001, standard deviation 1.002, within 1 of the mean: 68%
shuffled: ["four", "one", "three", "two", "five"]
same seed again: 0.0010
```

The normal numbers have mean close to 0, standard deviation close to 1, and about 68% of them fall within 1 of the mean, as the bell curve predicts. And a new generator with the same seed starts the same sequence again.

Chapter 9

The model

We now have all the tools to build the network itself. This chapter sets up its parameters; the next one runs it.

9.1 Embeddings: letters as vectors

How do we feed a letter to a layer? A layer wants a vector of numbers. We could feed the id itself, say 16 for `p`, but that would suggest that `p` (16) is "more" than `a` (1), and close to `o` (15), which is meaningless.

Instead, every symbol gets its own vector of 8 numbers, called its *embedding*. The 27 embeddings are the rows of a 27×8 matrix C. To feed a letter to the network, we look up its row: the letter with id s becomes row s of C, written C_s.

The crucial point is that the embeddings are *learned*. They start out random, and training adjusts them like every other weight. Letters that behave alike (say, the vowels) tend to end up with similar vectors, because that helps the network make good predictions. Nobody tells it to do this.

Looking up a row is a matrix operation in disguise. Write the symbol as a *one-hot* vector $\mathbf{e}_s$ with 27 components, all 0 except a 1 in position s. Then

$$C^{\mathsf{T}}\mathbf{e}_s = C_s,$$

where C^{T}, the *transpose* of C, is C with rows and columns swapped: multiplying by a one-hot vector picks out one column of C^{T}, which is one row of C. So an embedding is just a layer without bias, applied to a one-hot input. Looking the row up directly gives the same answer without multiplying by 26 zeros.

The model looks at the last 4 symbols. Their 4 embeddings, joined end to end, make the input vector $\mathbf{x}$ with $4 \times 8 = 32$ components:

$$\mathbf{x} = (C_{s_1}, C_{s_2}, C_{s_3}, C_{s_4}) \in \mathbb{R}^{32}.$$

9.2 The whole network

The forward pass

For context symbols $s_1, \ldots, s_4$:

$\mathbf{x} = (C_{s_1}, C_{s_2}, C_{s_3}, C_{s_4})$	embeddings, $\mathbb{R}^{32}$
$\mathbf{a} = W_1\mathbf{x} + \mathbf{b}_1$	hidden layer, $\mathbb{R}^{64}$
$\mathbf{h} = \tanh(\mathbf{a})$	applied to each component
$\mathbf{z} = W_2\mathbf{h} + \mathbf{b}_2$	output layer, $\mathbb{R}^{27}$
$\mathbf{p} = \mathrm{softmax}(\mathbf{z})$	probabilities

The numbers that training adjusts are the *parameters*: C (27×8), W_1 (64×32), $\mathbf{b}_1$ (64), W_2 (27×64) and $\mathbf{b}_2$ (27). In total:

$$27 \cdot 8 + 64 \cdot 32 + 64 + 27 \cdot 64 + 27 = 216 + 2048 + 64 + 1728 + 27 = 4083.$$

The sizes 4 (context), 8 (embedding) and 64 (hidden neurons) are not learned; we choose them. Such choices are called *hyperparameters*.

9.3 Choosing the starting values

The starting weights are random, but their *size* matters. Consider one hidden neuron before tanh: $a_i = \sum_j W_{1,ij}\, x_j$ (the biases start at 0). Suppose each x_j has mean 0 and variance 1 (the embeddings are standard normal) and each weight is a normal number times a scale σ. For independent quantities with mean 0, variances add, and the variance of a product of two of them is the product of their variances. So

$$\mathrm{Var}(a_i) = \sum_{j=0}^{n-1} \mathrm{Var}(W_{1,ij})\,\mathrm{Var}(x_j) = n\,\sigma^2,$$

where $n = 32$ is the number of inputs. (The variance is the square of the standard deviation: a measure of spread.) If we used $\sigma = 1$, the a_i would have a standard deviation of $\sqrt{32} \approx 5.7$, so most neurons would start deep in the flat parts of tanh, where its derivative is nearly 0 and, as we will see, learning is very slow. With $\sigma = 1/\sqrt{n}$ the variance is exactly 1: the neurons start in the lively middle part of tanh.

For the output layer we want the opposite: very small scores, so that softmax gives every symbol about the same probability, $1/27$. That is an honest start: the untrained model knows nothing, so it should not have strong opinions. We use $\sigma = 0.1/\sqrt{64}$, which gives scores with a standard deviation of at most 0.1.

9.4 The code

The model file starts with the files it uses. It also interns two libraries, `data/Json` and `io/Path`, which we will need in Chapter 16 for saving the model to a file.

The constructor `make` takes the three hyperparameters and a seed, and sets up all the parameters as described. The fields are listed after it, and the invariant states their shapes, so a parameter matrix with the wrong number of rows is caught immediately.

New file `next_letter_model.nex`

```
-- next_letter_model.nex: a tiny neural network that predicts the next
-- letter of a word.
--
-- It reads the last few letters of a partly typed word and gives a
-- probability for each of the 27 symbols (a..z, and "." for "the word is
-- finished").
--
--   last 4 letters --> embeddings --> hidden layer --> output layer --> softmax
--     (ids)           4 x 8 = 32       64 neurons      27 logits       27 probs
--                     lookup in C    tanh(W1·x + b1)   W2·h + b2
```

```
intern Vector_Math
intern Vocabulary
intern Rng
intern data/Json
intern io/Path

class Next_Letter_Model
  create
    -- A fresh, untrained model with random weights.
    make(context, emb_dim, hidden, seed: Integer)
      require
        positive_sizes: context > 0 and emb_dim > 0 and hidden > 0
      do
        -- Hyperparameters: chosen by us, not learned.
        this.context := context
        this.emb_dim := emb_dim
        this.hidden := hidden

        -- Parameters: learned during training. They start out random.
        let rng: Rng := create Rng.make(seed)
        let n_in: Integer := context * emb_dim

        -- C: one embedding row per symbol.
        c := random_matrix(rng, Vocabulary.SIZE, emb_dim, 1.0)

        -- W1, b1: the hidden layer. Scaling by 1/sqrt(n_in) keeps W1·x
        -- small at the start, where tanh is not yet flat.
        w1 := random_matrix(rng, hidden, n_in, 1.0 / n_in.to_real.sqrt)
        b1 := zeros(hidden)

        -- W2, b2: the output layer. Tiny weights make every symbol start
        -- out about equally likely: an honest "I know nothing yet".
        w2 := random_matrix(rng, Vocabulary.SIZE, hidden, 0.1 / hidden.to_real.sqrt)
        b2 := zeros(Vocabulary.SIZE)
      end

  feature
    context: Integer   -- how many previous letters the model sees
    emb_dim: Integer   -- length of each letter's embedding vector
    hidden: Integer    -- number of neurons in the hidden layer

    c: Array[Array[Real]]    -- embeddings,   Vocabulary.SIZE x emb_dim
    w1: Array[Array[Real]]   -- hidden layer, hidden x (context * emb_dim)
    b1: Array[Real]          --               hidden
    w2: Array[Array[Real]]   -- output layer, Vocabulary.SIZE x hidden
    b2: Array[Real]          --               Vocabulary.SIZE

  invariant
    embedding_rows: c.length = Vocabulary.SIZE
    hidden_rows: w1.length = hidden and b1.length = hidden
    output_rows: w2.length = Vocabulary.SIZE and b2.length = Vocabulary.SIZE
end
```

In the constructor, a parameter has the same name as a field (`context`). Inside the constructor, `context` means the parameter, and `this.context` means the field.

`random_matrix` makes a matrix of normal numbers, each multiplied by a scale:

next_letter_model.nex — add at the end

```
-- A rows x cols matrix of normal random numbers, each times `scale`.
function random_matrix(rng: Rng, rows, cols: Integer, scale: Real): Array[Array[Real]]
  do
    repeat rows do
      let row: Array[Real] := []
      repeat cols do
        row.add(rng.gauss() * scale)
      end
      result.add(row)
    end
  ensure
    right_rows: result.length = rows
  end
```

Two small methods complete the setup. The first turns the end of a typed word into the 4 ids the network reads. We pad the word with `context` full stops on the left, and then take its last `context` characters: `"app"` becomes `"...app"`, whose last four characters are `".app"`, which is `[0, 1, 16, 16]`.

next_letter_model.nex — add before the invariant

```
    -- The last `context` letters of a prefix as ids, padded with ".".
    --   "app" with context 4  ->  ".app"  ->  [0, 1, 16, 16]
    context_ids(prefix: String): Array[Integer]
      require
        known_symbols: is_lowercase_word(prefix) or prefix = ""
      do
        let padded: String := prefix
        repeat context do
          padded := Vocabulary.END + padded
        end
        let tail: String := padded.substring(padded.length - context, padded.length)
        across tail as ch do
          result.add(symbol_id(ch.to_string))
        end
      ensure
        full_context: result.length = context
      end

    num_parameters(): Integer do
      result := Vocabulary.SIZE * emb_dim
        + hidden * context * emb_dim + hidden
        + Vocabulary.SIZE * hidden + Vocabulary.SIZE
    end
```

Try it try_model.nex

```
intern Next_Letter_Model
intern Helpers

let con: Console := create Console
let model: Next_Letter_Model := create Next_Letter_Model.make(4, 8, 64, 42)
con.print_line("parameters: " + model.num_parameters.to_string)
con.print_line("context of \"app\":    " + model.context_ids("app").to_string)
```

```
con.print_line("context of \"banana\": " + model.context_ids("banana").to_string)
con.print_line("context of \"\":       " + model.context_ids("").to_string)
con.print_line("embedding of \"a\": " + vector_text(model.c.get(1), 2))
```

```
$ nex try_model.nex

parameters: 4083
context of "app":    [0, 1, 16, 16]
context of "banana": [14, 1, 14, 1]
context of "":       [0, 0, 0, 0]
embedding of "a": [1.52, -0.28, 0.47, 0.33, 0.97, -2.39, 1.26, 0.03]
```

The embedding of a is, for now, eight random numbers.

Chapter 10

The forward pass

Running the network on an input, from the embeddings to the probabilities, is called the *forward pass*. Training will need more than the final probabilities: to work out how to improve the weights, it needs the in-between values **x** and **h** too. So the forward pass returns all of them, in a small object:

next_letter_model.nex – add before the model class

```
-- What one forward pass computed. Backpropagation reuses these values
-- instead of computing them again.
class Forward_Pass
  create
    make(ids: Array[Integer], x, h, probs: Array[Real]) do
      this.ids := ids
      this.x := x
      this.h := h
      this.probs := probs
    end
  feature
    ids: Array[Integer]   -- the context letters, as symbol ids
    x: Array[Real]        -- their embeddings, joined into one vector
    h: Array[Real]        -- the hidden layer's outputs
    probs: Array[Real]    -- one probability per symbol
end
```

The method itself follows the box in the previous chapter line by line: look up and join the embeddings, run the hidden layer and tanh, run the output layer and softmax.

next_letter_model.nex – add before num_parameters

```
    -- Run the network on one context.
    forward(ids: Array[Integer]): Forward_Pass
      require
        full_context: ids.length = context
      do
        -- 1. Embedding lookup: replace each id by its row of C, and join
        --    the rows into one input vector x.
        let x: Array[Real] := []
        across ids as id do
          across c.get(id) as v do
            x.add(v)
          end
        end

        -- 2. Hidden layer: h = tanh(W1·x + b1).
        let h: Array[Real] := []
        across affine(w1, x, b1) as v do
          h.add(tanh(v))
        end
```

```
    -- 3. Output layer: one score per symbol, then softmax.
    let probs: Array[Real] := softmax(affine(w2, h, b2))

    result := create Forward_Pass.make(ids, x, h, probs)
  ensure
    one_probability_per_symbol: result.probs.length = Vocabulary.SIZE
  end
```

Try it `try_forward.nex`

```
intern Next_Letter_Model
intern Helpers

let con: Console := create Console
let model: Next_Letter_Model := create Next_Letter_Model.make(4, 8, 64, 42)
let fp: Forward_Pass := model.forward(model.context_ids("app"))
con.print_line("x has " + fp.x.length.to_string + " numbers, h has "
  + fp.h.length.to_string + ", probs has " + fp.probs.length.to_string)
con.print_line("first hidden values: " + vector_text(fp.h.take(6), 3))
con.print_line("probabilities after \"app\":")
from
  let i: Integer := 0
until
  i = Vocabulary.SIZE
do
  con.print(symbol_at(i) + " " + pad_left(percent(fp.probs.get(i)), 3) + "   ")
  if i % 9 = 8 then
    con.print_line("")
  end
  i := i + 1
end
con.print_line("they add up to " + fixed(sum(fp.probs), 6))
```

```
$ nex try_forward.nex

x has 32 numbers, h has 64, probs has 27
first hidden values: [-0.756, 0.399, 0.278, 0.894, -0.696, 0.471]
probabilities after "app":
.  4%   a  4%   b  3%   c  4%   d  4%   e  4%   f  4%   g  3%   h  4%
i  4%   j  4%   k  4%   l  3%   m  4%   n  4%   o  4%   p  4%   q  4%
r  4%   s  4%   t  4%   u  3%   v  3%   w  4%   x  4%   y  3%   z  4%
they add up to 1.000000
```

The untrained model gives every symbol close to $1/27 \approx 3.7\%$, as we designed it to. The hidden values, on the other hand, are well spread between -1 and 1: the neurons are in the lively part of tanh.

Chapter 11

Training data and the loss

To train the model we need examples of right answers, and a way to score how wrong the model is.

11.1 Examples for free

Nobody has to label anything by hand: every word labels itself. In `apple`, the letter after each prefix is the right answer for that prefix. With the padding and the end marker, `apple` gives six examples:

context	ids	next symbol
`....`	0, 0, 0, 0	`a`
`...a`	0, 0, 0, 1	`p`
`..ap`	0, 0, 1, 16	`p`
`.app`	0, 1, 16, 16	`l`
`appl`	1, 16, 16, 12	`e`
`pple`	16, 16, 12, 5	`.` (the word ends)

A word of n letters gives $n + 1$ examples. Learning from data whose answers come from the data itself is called *self-supervised* learning, and it is how language models learn too: every piece of text supplies its own "next piece".

New file `training_data.nex`

```
-- training_data.nex: turning a word list into training examples, and
-- measuring how well a model does on them.
--
-- Nobody labels anything by hand. Every word labels itself: in "apple",
-- the letter after each prefix is the right answer for that prefix.
--
--   context "...." -> "a"
--   context "...a" -> "p"
--   context "..ap" -> "p"
--   context ".app" -> "l"
--   context "appl" -> "e"
--   context "pple" -> "."   <- the word ends here

intern Vocabulary
intern Vector_Math
intern Next_Letter_Model
intern io/Path

class Training_Data
  feature
    -- Keeps -log(p) finite even if a probability rounds to exactly 0.
    TINY = 1.0e-12
end
```

```
-- One training example: a context and the symbol that follows it.
class Example
  create
    make(ids: Array[Integer], target: Integer) do
      this.ids := ids
      this.target := target
    end
  feature
    ids: Array[Integer]
    target: Integer
end
```

Reading the word list: split the file into lines, keep the plain lowercase words of two letters or more, and drop repeats by collecting them in a `Set`. Sorting the result makes the order independent of how the file happened to be written, so that the same seed gives the same training run.

`training_data.nex` – add at the end

```
-- The plain lowercase words (a..z only, 2 letters or more) in a file with
-- one word per line, without repeats, in sorted order. Proper nouns
-- ("Paris"), possessives ("apple's") and accented words are skipped, so
-- the vocabulary stays at exactly 27 symbols.
function load_words(path: String): Array[String] do
  let keep: Set[String] := #{}
  across create Path.make(path).read_text().split("\n") as line do
    let w: String := line.trim()
    if w.length >= 2 and is_lowercase_word(w) then
      keep.add(w)
    end
  end
  result := keep.to_array().sort()
end
```

Making the examples follows the table: pad the word with `context` full stops in front and one behind, then slide a window of `context` symbols along it. For `apple` the padded word is `....apple.`; the window at position i is the context, and the symbol just after the window is the answer.

`training_data.nex` – add at the end

```
-- Every (context, next symbol) pair in every word, as shown at the top.
function make_examples(words: Array[String], context: Integer): Array[Example]
  require
    positive_context: context > 0
  do
    let pad: String := ""
    repeat context do
      pad := pad + Vocabulary.END
    end
    across words as word do
      let padded: String := pad + word + Vocabulary.END
      from
        let i: Integer := 0
```

```
      until
        i = word.length + 1
      do
        let ids: Array[Integer] := []
        across padded.substring(i, i + context) as ch do
          ids.add(symbol_id(ch.to_string))
        end
        let target: Integer := symbol_id(padded.substring(i + context, i + context + 1))
        result.add(create Example.make(ids, target))
        i := i + 1
      end
    end
  end
```

11.2 Measuring wrongness: cross-entropy

For one example, the model gives a probability p_t to the right answer t (the *target*). A good model gives the right answer a high probability. The standard way to score this is the *cross-entropy loss*:

Cross-entropy loss

$$L = -\ln p_t$$

where p_t is the probability the model gives to the right answer. Lower is better. Over many examples, we take the average.

ln is the *natural logarithm*, the inverse of the exponential: $\ln e^v = v$. For numbers between 0 and 1 it is negative, so the minus sign makes the loss positive. A few values show how it behaves:

p_t	$-\ln p_t$
1.0	0.000
0.5	0.693
$1/27 \approx 0.037$	3.296
0.01	4.605
0.0001	9.210

A perfect, certain prediction costs nothing. Each halving of the probability adds the same amount, $\ln 2 \approx 0.69$, to the loss. And a confident wrong answer (a tiny p_t) is punished very hard.

The value $\ln 27 \approx 3.296$ is our yardstick: it is the loss of a model that knows nothing and gives every symbol $1/27$. Any model with a lower loss has learned something.

There is a nice way to read a loss value: e^L is the number of equally likely choices the model is, in effect, choosing among (this is called the *perplexity*). The know-nothing model has $e^{3.296} = 27$; a model with loss 2.1 is about as uncertain as a choice among $e^{2.1} \approx 8$ letters.

Probabilities from softmax are never exactly 0 in theory, but a very small one can round to 0, and $\ln 0 = -\infty$. Adding a tiny number, 10^{-12}, keeps the loss finite without changing it noticeably.

`training_data.nex` — add at the end

```
-- The loss of one prediction: -log(probability given to the right answer).
function cross_entropy(probs: Array[Real], target: Integer): Real do
  result := -(probs.get(target) + Training_Data.TINY).log
end
```

Finally, a way to measure a model on many examples: the average loss, and the *accuracy*, which is how often the model's top choice is the right answer.

`training_data.nex` — add at the end

```
-- Average loss and accuracy over a set of examples.
class Evaluation
  create
    make(loss, accuracy: Real) do
      this.loss := loss
      this.accuracy := accuracy
    end
  feature
    loss: Real       -- average cross-entropy: lower is better
    accuracy: Real   -- fraction of examples where the top guess was right
end

-- Average cross-entropy and top-1 accuracy of a model on some examples.
function evaluate(model: Next_Letter_Model, examples: Array[Example]): Evaluation
  require
    not_empty: examples.length > 0
  do
    let total_loss: Real := 0.0
    let correct: Integer := 0
    across examples as ex do
      let probs: Array[Real] := model.forward(ex.ids).probs
      total_loss := total_loss + cross_entropy(probs, ex.target)
      if argmax(probs) = ex.target then
        correct := correct + 1
      end
    end
    let n: Real := examples.length.to_real
    result := create Evaluation.make(total_loss / n, correct.to_real / n)
  end
```

Try it `try_examples.nex`

```
intern Training_Data
intern Next_Letter_Model
intern Helpers

let con: Console := create Console
across make_examples(["apple"], 4) as ex do
  let context: String := ""
  across ex.ids as id do
    context := context + symbol_at(id)
  end
  con.print_line(context + " -> " + symbol_at(ex.target))
end
```

```
let words: Array[String] := load_words("/usr/share/dict/words")
con.print_line(words.length.to_string + " usable words, from "
  + words.get(0) + " to " + words.get(words.length - 1))
let examples: Array[Example] := make_examples(words.take(2000), 4)
let model: Next_Letter_Model := create Next_Letter_Model.make(4, 8, 64, 42)
let e: Evaluation := evaluate(model, examples)
con.print_line("untrained model on " + examples.length.to_string + " examples: loss "
  + fixed(e.loss, 3) + ", accuracy " + percent(e.accuracy))
con.print_line("knowing nothing would give: loss " + fixed(27.0.log, 3)
  + ", accuracy about " + percent(1.0 / 27.0))
```

```
$ nex try_examples.nex

.... -> a
...a -> p
..ap -> p
.app -> l
appl -> e
pple -> .
63849 usable words, from aardvark to zygotes
untrained model on 18971 examples: loss 3.297, accuracy 1%
knowing nothing would give: loss 3.296, accuracy about 4%
```

Your word count will differ if your word list is different. The untrained model scores close to the know-nothing loss, as expected. Now we need to make it learn.

Chapter 12

Learning by going downhill

The loss depends on all 4083 parameters. Training means finding parameter values that make the average loss small. We do this by repeatedly changing every parameter a little in the direction that makes the loss go down. To know that direction, we need derivatives.

12.1 Derivatives

The *derivative* of a function f at a point w, written $f'(w)$ or $\frac{df}{dw}$, is the slope of its graph there: how much f changes per unit change of w, for very small changes,

$$f'(w) \approx \frac{f(w+\varepsilon) - f(w)}{\varepsilon} \qquad \text{for a tiny } \varepsilon.$$

A few rules are all we need: the derivative of w^2 is $2w$, of a constant is 0, of e^w is e^w, of $\ln w$ is $1/w$, and derivatives of sums are sums of derivatives.

When a function depends on many variables, like our loss, its *partial derivative* $\frac{\partial L}{\partial w}$ with respect to one of them is the ordinary derivative when all the others are held fixed. The list of all partial derivatives, one per parameter, is the *gradient*. It points in the direction in which the loss increases fastest. So the loss decreases fastest in the opposite direction.

12.2 Gradient descent

Gradient descent

For every parameter w, repeatedly:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

The step size $\eta > 0$ (the Greek letter eta) is the *learning rate*.

If the slope $\frac{\partial L}{\partial w}$ is positive, increasing w increases the loss, so we decrease w, and vice versa. Steep slopes give big steps, and near the bottom, where the slope flattens out, the steps get small.

Let's watch this on the smallest possible example, a "loss" with one parameter: $L(w) = (w-3)^2$. Its lowest point is obviously at $w = 3$, and its derivative is $L'(w) = 2(w-3)$.

Try it `try_descent.nex`

```
intern Helpers

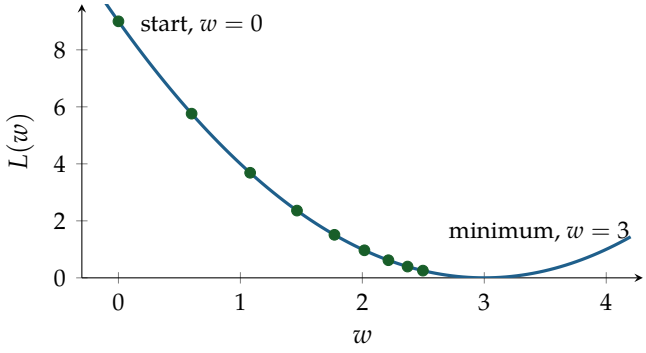
-- Minimize L(w) = (w - 3)^2, starting from w = 0.
function descend(lr: Real) do
  let con: Console := create Console
  con.print_line("learning rate " + fixed(lr, 2) + ":")
  let w: Real := 0.0
```

```
  from
    let step: Integer := 0
  until
    step > 12
  do
    let loss: Real := (w - 3.0) * (w - 3.0)
    let slope: Real := 2.0 * (w - 3.0)
    if step % 3 = 0 then
      con.print_line("  step " + pad_left(step.to_string, 2) + ": w = "
        + pad_left(fixed(w, 4), 9) + "   loss = " + fixed(loss, 4))
    end
    w := w - lr * slope
    step := step + 1
  end
end

descend(0.1)
descend(0.45)
descend(1.1)
```

```
$ nex try_descent.nex

learning rate 0.10:
  step  0: w =    0.0000   loss = 9.0000
  step  3: w =    1.4640   loss = 2.3593
  step  6: w =    2.2136   loss = 0.6185
  step  9: w =    2.5973   loss = 0.1621
  step 12: w =    2.7938   loss = 0.0425
learning rate 0.45:
  step  0: w =    0.0000   loss = 9.0000
  step  3: w =    2.9970   loss = 0.0000
  step  6: w =    3.0000   loss = 0.0000
  step  9: w =    3.0000   loss = 0.0000
  step 12: w =    3.0000   loss = 0.0000
learning rate 1.10:
  step  0: w =    0.0000   loss = 9.0000
  step  3: w =    8.1840   loss = 26.8739
  step  6: w =   -5.9580   loss = 80.2449
  step  9: w =   18.4793   loss = 239.6100
  step 12: w =  -23.7483   loss = 715.4716
```



With learning rate 0.1, w creeps toward 3 (the dots in the plot), and the loss falls steadily. With 0.45 it gets there much faster. With 1.1, the steps overshoot: w jumps from one side of the valley to the other, further each time, and the loss explodes.

We can see exactly why. One step changes w to $w - \eta \cdot 2(w-3)$, so the distance from the minimum changes as

$$(w-3) \;\leftarrow\; (1-2\eta)\,(w-3).$$

Every step multiplies the distance by $1-2\eta$: by 0.8 for $\eta = 0.1$, by 0.1 for $\eta = 0.45$, and by -1.2 for $\eta = 1.1$, which flips the sign and grows. Gradient descent converges here only if $|1-2\eta| < 1$, that is, $0 < \eta < 1$. Real losses are not this simple, but the lesson carries over: too small a learning rate is slow, too large is unstable, and finding a good one takes some experiment.

12.3 Many examples: stochastic gradient descent

Our real loss is an average over half a million examples. Computing its exact gradient would mean running all of them through the network for every single step. Instead, each step uses a small random *minibatch* of 32 examples, and the average of their gradients, which is a noisy but useful estimate of the true one. (The gradient of an average is the average of the gradients, because derivatives of sums are sums of derivatives.) This is *stochastic gradient descent*: "stochastic" means "random". Many cheap, noisy steps get much further than a few exact, expensive ones.

12.4 The chain rule

One question remains: how to compute $\frac{\partial L}{\partial w}$ for a weight deep inside the network. The loss doesn't depend on such a weight directly, but through a chain of steps: the weight affects a neuron, which affects the scores, which affect the probabilities, which affect the loss. The *chain rule* says how derivatives combine along such a chain:

Chain rule

If L depends on y, and y depends on w, then

$$\frac{dL}{dw} = \frac{dL}{dy} \cdot \frac{dy}{dw}.$$

If w affects L through several in-between values $y_0, y_1, \ldots$, the contributions of all the paths add up:

$$\frac{\partial L}{\partial w} = \sum_i \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial w}.$$

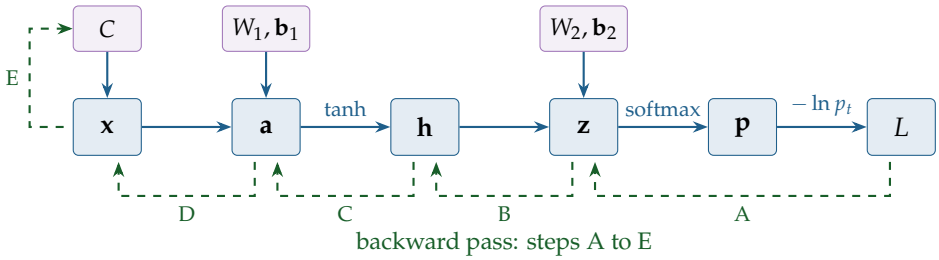
In words: the effect of a small change in w on L is the effect of w on y, times the effect of y on L. If changing w by 0.01 changes y by 0.03 ($dy/dw = 3$), and changing y by 0.03 changes L by 0.06 ($dL/dy = 2$), then changing w by 0.01 changes L by 0.06, and $dL/dw = 6 = 2 \cdot 3$.

Applying the chain rule backwards through the network, one layer at a time, is called *backpropagation*. It is the subject of the next chapter.

Chapter 13

Backpropagation

We now work out, for one example, the partial derivative of the loss with respect to every parameter. The plan is to go backwards through the forward pass, from the loss to the input. At each stage we already know the derivatives with respect to that stage's outputs, and we use the chain rule to get the derivatives with respect to its inputs and its parameters.



A word on notation: for any value v in the network we write $\frac{\partial L}{\partial v}$, "how much the loss changes per unit change of v". In the code, the same quantity is called `dv`: `dz`, `dh` and so on. For a vector, it is a vector of the same length, one partial derivative per component.

13.1 The gradients container

Each example produces a gradient for every parameter. We add the gradients of all examples in a minibatch together, and use their average for the update. The container has one zero-filled matrix or vector per parameter, with the same shape:

`next_letter_model.nex` – add before the model class

```
-- Zero-filled containers with the same shapes as the model's parameters.
-- During a minibatch, each example adds its gradients here.
class Gradients
  create
    make(model: Next_Letter_Model) do
      c := zero_matrix(Vocabulary.SIZE, model.emb_dim)
      w1 := zero_matrix(model.hidden, model.context * model.emb_dim)
      b1 := zeros(model.hidden)
      w2 := zero_matrix(Vocabulary.SIZE, model.hidden)
      b2 := zeros(Vocabulary.SIZE)
    end
  feature
    c: Array[Array[Real]]
    w1: Array[Array[Real]]
    b1: Array[Real]
    w2: Array[Array[Real]]
    b2: Array[Real]
end
```

13.2 Step A: softmax and cross-entropy

The loss is $L = -\ln p_t$ with $p_t = e^{z_t}/S$, where $S = \sum_j e^{z_j}$. The logarithm of a quotient is a difference of logarithms, and $\ln e^{z_t} = z_t$, so

$$L = -\ln \frac{e^{z_t}}{S} = -z_t + \ln S.$$

Now take the partial derivative with respect to one score z_i. The first term, $-z_t$, gives -1 if $i = t$, and 0 otherwise. For the second, the chain rule with $(\ln S)' = 1/S$ gives

$$\frac{\partial \ln S}{\partial z_i} = \frac{1}{S}\frac{\partial S}{\partial z_i} = \frac{e^{z_i}}{S} = p_i,$$

because only the term e^{z_i} of the sum S depends on z_i. So:

Step A

$$\frac{\partial L}{\partial z_i} = p_i - y_i, \qquad \text{where } y_i = \begin{cases} 1 & \text{if } i = t \\ 0 & \text{otherwise.} \end{cases}$$

This is a beautifully simple result: the gradient is "what the model said" minus "what it should have said". For example, if the model gives probabilities $(0.7, 0.2, 0.1)$ and the right answer is the second symbol, the gradient is $(0.7, -0.8, 0.1)$. Gradient descent *subtracts* the gradient, so the right answer's score goes up, and every wrong answer's score goes down in proportion to how much probability it wrongly took. And the gradient is small when the model is already nearly right.

In code: copy the probabilities, and subtract 1 at the target. This is the start of the `backward` method; the next boxes continue it.

`next_letter_model.nex` – add before num_parameters

```
    -- Add one example's gradients into g, by the chain rule, from the
    -- loss back to the input:
    --   loss -> logits -> W2, b2, h -> tanh -> W1, b1, x -> C
    backward(fp: Forward_Pass, target: Integer, g: Gradients)
      require
        valid_target: target >= 0 and target < Vocabulary.SIZE
      do
        -- Step A: softmax followed by cross-entropy has a very simple
        -- gradient with respect to the logits:
        --     dL/dz[i] = p[i] - (1 if i is the target, else 0)
        let dz: Array[Real] := fp.probs.clone()
        dz.set(target, dz.get(target) - 1.0)
```

13.3 Step B: the output layer

The scores are $z_i = \sum_j W_{2,ij} h_j + b_{2,i}$. We know $\frac{\partial L}{\partial z_i}$ from step A, and we want the derivatives with respect to W_2, $\mathbf{b}_2$ and $\mathbf{h}$.

A weight $W_{2,ij}$ affects only one score, z_i, and $\frac{\partial z_i}{\partial W_{2,ij}} = h_j$. A bias $b_{2,i}$ also affects only z_i, with $\frac{\partial z_i}{\partial b_{2,i}} = 1$. But a hidden value h_j affects *every* score, through the weights $W_{2,ij}$, so by the chain rule its contributions add up over all the scores:

Step B, and the same for any layer $\mathbf{z} = W\mathbf{h} + \mathbf{b}$

$$\frac{\partial L}{\partial W_{2,ij}} = \frac{\partial L}{\partial z_i} h_j, \qquad \frac{\partial L}{\partial b_{2,i}} = \frac{\partial L}{\partial z_i}, \qquad \frac{\partial L}{\partial h_j} = \sum_i \frac{\partial L}{\partial z_i} W_{2,ij}.$$

Look at these one row i at a time. Row i of the gradient of W_2 is the vector $\mathbf{h}$ times the number $\frac{\partial L}{\partial z_i}$. And $\frac{\partial L}{\partial \mathbf{h}}$ is the sum, over all rows i, of row i of W_2 times $\frac{\partial L}{\partial z_i}$. Both are exactly what `add_scaled` does:

`next_letter_model.nex` – continued

```
        -- Step B: output layer, z = W2·h + b2.
        --     dW2[i][j] += dz[i] * h[j]     db2[i] += dz[i]
        --     dh[j]     += dz[i] * W2[i][j]
        let dh: Array[Real] := zeros(hidden)
        from
          let i: Integer := 0
        until
          i = Vocabulary.SIZE
        do
          add_scaled(g.w2.get(i), dz.get(i), fp.h)
          g.b2.set(i, g.b2.get(i) + dz.get(i))
          add_scaled(dh, dz.get(i), w2.get(i))
          i := i + 1
        end
```

Note that we *add* into `g`, rather than overwrite it, so that the gradients of all the examples in a minibatch accumulate.

13.4 Step C: through tanh

Each hidden value is $h_j = \tanh(a_j)$, and depends only on its own a_j. By the chain rule and the derivative of tanh from Chapter 6:

Step C

$$\frac{\partial L}{\partial a_j} = \frac{\partial L}{\partial h_j} \tanh'(a_j) = \frac{\partial L}{\partial h_j} (1 - h_j^2).$$

We kept $\mathbf{h}$ from the forward pass, so this is cheap. It also shows why the starting weights mattered: if a neuron is deep in a flat part of tanh, h_j is close to ± 1, the factor $1 - h_j^2$ is close to 0, and almost no gradient gets through to the layer below. That neuron hardly learns.

`next_letter_model.nex` – continued

```
        -- Step C: through tanh. tanh'(a) = 1 - tanh(a)^2 = 1 - h^2, and
        -- we already have h, so this costs almost nothing.
        let da: Array[Real] := []
        from
          let j: Integer := 0
        until
          j = hidden
        do
          let hj: Real := fp.h.get(j)
```

```
      da.add(dh.get(j) * (1.0 - hj * hj))
      j := j + 1
    end
```

13.5 Step D: the hidden layer

The hidden layer is $\mathbf{a} = W_1\mathbf{x} + \mathbf{b}_1$: the same kind of layer as in step B, so the same three formulas apply, with W_1, $\mathbf{b}_1$, $\mathbf{x}$ and $\frac{\partial L}{\partial \mathbf{a}}$ in place of W_2, $\mathbf{b}_2$, $\mathbf{h}$ and $\frac{\partial L}{\partial \mathbf{z}}$:

$$\frac{\partial L}{\partial W_{1,ij}} = \frac{\partial L}{\partial a_i}\, x_j, \qquad \frac{\partial L}{\partial b_{1,i}} = \frac{\partial L}{\partial a_i}, \qquad \frac{\partial L}{\partial x_j} = \sum_i \frac{\partial L}{\partial a_i}\, W_{1,ij}.$$

next_letter_model.nex – continued

```
    -- Step D: hidden layer, a = W1·x + b1. Same pattern as step B.
    let dx: Array[Real] := zeros(fp.x.length)
    from
      let i: Integer := 0
    until
      i = hidden
    do
      add_scaled(g.w1.get(i), da.get(i), fp.x)
      g.b1.set(i, g.b1.get(i) + da.get(i))
      add_scaled(dx, da.get(i), w1.get(i))
      i := i + 1
    end
```

13.6 Step E: the embeddings

The input $\mathbf{x}$ was the embeddings of the 4 context symbols, joined: components 0–7 are the row C_{s_1}, components 8–15 are C_{s_2}, and so on. In general, component $k \cdot 8 + m$ of $\mathbf{x}$ is component m of the row C_{s_k} (counting k from 0). So the gradient of $\mathbf{x}$ is cut back into 4 pieces of 8, and each piece is added to the gradient of the row it came from:

Step E

$$\frac{\partial L}{\partial C_{s,m}} = \sum_{k\,:\,s_k = s} \frac{\partial L}{\partial x_{8k+m}}$$

The sum is over every position k where the symbol s appears in the context. In `.app`, the letter `p` appears twice, so its row of C gets the gradients of both positions. Rows of symbols that are not in the context get nothing from this example.

next_letter_model.nex – continued

```
    -- Step E: embeddings. x was the context letters' rows of C joined
    -- together, so cut dx back into pieces and add each piece to the
    -- row it came from. A letter that appears twice ("pp") gets both.
    from
      let k: Integer := 0
```

```
      until
        k = context
      do
        let row: Array[Real] := g.c.get(fp.ids.get(k))
        from
          let d: Integer := 0
        until
          d = emb_dim
        do
          row.set(d, row.get(d) + dx.get(k * emb_dim + d))
          d := d + 1
        end
        k := k + 1
      end
    end
```

The final `end` closes the method. That is the whole of backpropagation for our network: about fifty lines. Here are all the steps together, written with vectors:

The backward pass, in summary

$$\text{A:}\quad \frac{\partial L}{\partial \mathbf{z}} = \mathbf{p} - \mathbf{y}$$

$$\text{B:}\quad \frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial \mathbf{z}}\,\mathbf{h}^{\mathsf{T}}, \quad \frac{\partial L}{\partial \mathbf{b}_2} = \frac{\partial L}{\partial \mathbf{z}}, \quad \frac{\partial L}{\partial \mathbf{h}} = W_2^{\mathsf{T}}\frac{\partial L}{\partial \mathbf{z}}$$

$$\text{C:}\quad \frac{\partial L}{\partial \mathbf{a}} = \frac{\partial L}{\partial \mathbf{h}} \odot (1 - \mathbf{h} \odot \mathbf{h})$$

$$\text{D:}\quad \frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \mathbf{a}}\,\mathbf{x}^{\mathsf{T}}, \quad \frac{\partial L}{\partial \mathbf{b}_1} = \frac{\partial L}{\partial \mathbf{a}}, \quad \frac{\partial L}{\partial \mathbf{x}} = W_1^{\mathsf{T}}\frac{\partial L}{\partial \mathbf{a}}$$

$$\text{E:}\quad \frac{\partial L}{\partial C_s} \mathrel{+}= \text{the pieces of } \frac{\partial L}{\partial \mathbf{x}} \text{ at the positions of } s$$

Here $\odot$ multiplies vectors component by component, and $\mathbf{u}\mathbf{v}^{\mathsf{T}}$ is the matrix with entries $u_i v_j$.

But is it right? A mistake in a formula or an index here would not crash anything. The network would simply learn badly, or not at all, and it would be very hard to tell why. The next chapter builds a test that catches such mistakes.

Chapter 14

Checking the gradients

We derived the gradients with the chain rule. We can also *measure* them, directly from the definition of the derivative: nudge one parameter up a tiny bit, run the network, nudge it down, run it again, and see how much the loss changed. If the measured and the derived gradients agree, the backward pass is right.

14.1 The math: central differences

The most accurate simple way to measure a derivative is the *central difference*:

Central difference

$$f'(w) \approx \frac{f(w+\varepsilon) - f(w-\varepsilon)}{2\varepsilon}$$

with an error proportional to ε^2.

Why ε^2? Taylor's theorem says that near w,

$$f(w \pm \varepsilon) = f(w) \pm \varepsilon f'(w) + \tfrac{1}{2}\varepsilon^2 f''(w) \pm \tfrac{1}{6}\varepsilon^3 f'''(w) + \dots$$

Subtracting the two versions, the $f(w)$ and ε^2 terms cancel:

$$f(w+\varepsilon) - f(w-\varepsilon) = 2\varepsilon f'(w) + \tfrac{1}{3}\varepsilon^3 f'''(w) + \dots$$

and dividing by 2ε leaves $f'(w)$ plus an error of order ε^2. With $\varepsilon = 10^{-5}$, that error is around 10^{-10}, far smaller than any mistake we are looking for. (A one-sided difference, $(f(w+\varepsilon) - f(w))/\varepsilon$, would only be accurate to order ε.)

Measuring every gradient this way would need two forward passes per parameter, 8166 for one example, which is why we don't train like this. But it is perfect for spot-checking a few parameters of each kind.

14.2 A file of checks

We put the gradient check, and checks for everything else we have built, in one file that we can run at any time. First, a small class that counts passes and failures:

New file `checks.nex`

```
-- checks.nex: executable checks for every piece of the project.
-- Run: nex checks.nex
--
-- The most important one is the gradient check. It compares the gradients
-- that backward computes with the chain rule against gradients measured
-- directly, by nudging one parameter a tiny bit and watching the loss.
-- If backpropagation has a bug, the two disagree.
```

```
intern Vector_Math
intern Vocabulary
intern Rng
intern Next_Letter_Model
intern Training_Data
intern Helpers
intern io/Path

class Checker
  create
    make() do
      con := create Console
    end
  feature
    con: Console
    passed: Integer
    failed: Integer

    check(label: String, ok: Boolean) do
      if ok then
        passed := passed + 1
        con.print_line("PASS " + label)
      else
        failed := failed + 1
        con.print_line("FAIL " + label)
      end
    end

    close_to(label: String, expected, actual, tolerance: Real) do
      check(label + " (expected " + expected.to_string
          + ", got " + actual.to_string + ")",
        (expected - actual).abs <= tolerance)
    end

    summary() do
      con.print_line("---")
      con.print_line(passed.to_string + " passed, " + failed.to_string + " failed")
      if failed > 0 then
        exit(1)
      end
    end
end

let c: Checker := create Checker.make()

c.summary()
```

Every check we add goes between `let c` and `c.summary()`. `summary` exits with status 1 if anything failed, so the checks can also be run automatically.

Let's start by locking in what we built in the earlier chapters. These checks repeat a few of the try-it results, so that if a later change breaks something, we find out at once:

`checks.nex` – add before the summary

```
-- Vector math
c.close_to("dot", 32.0, dot([1.0, 2.0, 3.0], [4.0, 5.0, 6.0]), 0.0)
```

```
let a: Array[Real] := affine([[1.0, 0.0], [1.0, 1.0]], [2.0, 3.0], [0.5, -1.0])
c.check("affine", a.get(0) = 2.5 and a.get(1) = 4.0)
let s: Array[Real] := softmax([1.0, 2.0, 3.0])
c.close_to("softmax sums to 1", 1.0, sum(s), 1.0e-12)
c.close_to("softmax[2]", 0.6652409557748219, s.get(2), 1.0e-12)
c.close_to("softmax of huge scores", 0.5, softmax([1000.0, 1000.0]).get(0), 1.0e-12)
c.close_to("tanh(0.5)", 0.46211715726000974, tanh(0.5), 1.0e-12)
c.close_to("tanh(100)", 1.0, tanh(100.0), 0.0)
c.close_to("tanh(-100)", -1.0, tanh(-100.0), 0.0)
c.check("argmax", argmax([0.1, 0.7, 0.2]) = 1)
c.check("fixed", fixed(2.5, 3) = "2.500" and fixed(-0.0499, 2) = "-0.05"
  and fixed(3.14159, 0) = "3")
c.check("percent", percent(0.257) = "26%")
```

checks.nex — add before the summary

```
-- Random numbers
let rng: Rng := create Rng.make(42)
let mean: Real := 0.0
let square: Real := 0.0
repeat 20000 do
  let z: Real := rng.gauss()
  mean := mean + z / 20000.0
  square := square + z * z / 20000.0
end
c.close_to("gauss mean", 0.0, mean, 0.03)
c.close_to("gauss variance", 1.0, square - mean * mean, 0.05)
let r1: Rng := create Rng.make(7)
let r2: Rng := create Rng.make(7)
c.check("same seed, same numbers", r1.uniform() = r2.uniform())

-- Vocabulary
c.check("symbol_id", symbol_id(".") = 0 and symbol_id("a") = 1 and symbol_id("z") = 26)
c.check("symbol_at", symbol_at(16) = "p")
c.check("is_lowercase_word", is_lowercase_word("apple") and not is_lowercase_word("Apple")
  and not is_lowercase_word("it's"))
```

checks.nex — add before the summary

```
-- Training examples
let examples: Array[Example] := make_examples(["apple"], 4)
c.check("apple gives 6 examples", examples.length = 6)
c.check("first example", examples.get(0).ids = [0, 0, 0, 0]
  and examples.get(0).target = 1)
c.check("last example ends the word", examples.get(5).ids = [16, 16, 12, 5]
  and examples.get(5).target = 0)

-- The model
let model: Next_Letter_Model := create Next_Letter_Model.make(4, 8, 64, 42)
c.check("4083 parameters", model.num_parameters = 4083)
c.check("context_ids", model.context_ids("app") = [0, 1, 16, 16])
c.check("context_ids keeps the last letters",
  model.context_ids("banana") = [14, 1, 14, 1])
c.close_to("untrained loss is near ln 27", 27.0.log, evaluate(model, examples).loss, 0.1)
```

14.3 The gradient check

Now the gradient check itself. `loss_on` runs the model on one example and returns its loss. `gradient_matches` takes one parameter, given as a row of a parameter matrix and a column in it, nudges it by $\pm\varepsilon$, measures the central difference, puts the parameter back the way it was, and compares the measurement with the gradient that `backward` computed.

How close must they be? The measured value has small rounding errors of its own, so we allow a difference of 10^{-6}, plus 10^{-4} of the gradient's size for large gradients. A mistake in the backward pass, such as a missing factor or a wrong index, typically gives a difference of 10% or more.

`checks.nex` – add before the line let c

```
-- The loss of a model on one example.
function loss_on(model: Next_Letter_Model, ex: Example): Real do
  result := cross_entropy(model.forward(ex.ids).probs, ex.target)
end

-- Compare backward's gradient for parameter row[col] with a measured one:
--     dL/dp  is about  (L(p + eps) - L(p - eps)) / (2 eps)
function gradient_matches(model: Next_Letter_Model, ex: Example, row: Array[Real],
                          col: Integer, analytic: Real): Boolean do
  let eps: Real := 1.0e-5
  let saved: Real := row.get(col)
  row.set(col, saved + eps)
  let up: Real := loss_on(model, ex)
  row.set(col, saved - eps)
  let down: Real := loss_on(model, ex)
  row.set(col, saved)
  let numeric: Real := (up - down) / (2.0 * eps)
  result := (numeric - analytic).abs <= 1.0e-6 + 1.0e-4 * numeric.abs
end
```

The vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ are passed as the "row" themselves. For each parameter, we test a couple of entries. For C we test the row of `p`, which appears twice in the context `.app`, to check that both of its gradients are added up.

`checks.nex` – add before the summary

```
-- The gradient check, on one example, for a few parameters of every kind.
let ex: Example := examples.get(3)            -- ".app" -> "l"
let grads: Gradients := create Gradients.make(model)
model.backward(model.forward(ex.ids), ex.target, grads)
c.check("gradient of b2",
  gradient_matches(model, ex, model.b2, 12, grads.b2.get(12))
  and gradient_matches(model, ex, model.b2, 5, grads.b2.get(5)))
c.check("gradient of W2",
  gradient_matches(model, ex, model.w2.get(12), 3, grads.w2.get(12).get(3))
  and gradient_matches(model, ex, model.w2.get(0), 40, grads.w2.get(0).get(40)))
c.check("gradient of b1",
  gradient_matches(model, ex, model.b1, 7, grads.b1.get(7)))
c.check("gradient of W1",
  gradient_matches(model, ex, model.w1.get(7), 30, grads.w1.get(7).get(30))
  and gradient_matches(model, ex, model.w1.get(63), 0, grads.w1.get(63).get(0)))
-- "p" (id 16) appears twice in ".app", so its embedding gets two gradients.
c.check("gradient of C",
```

```
  gradient_matches(model, ex, model.c.get(16), 2, grads.c.get(16).get(2))
  and gradient_matches(model, ex, model.c.get(1), 5, grads.c.get(1).get(5)))
```

```
$ nex checks.nex

PASS dot (expected 32.0, got 32.0)
PASS affine
PASS softmax sums to 1 (expected 1.0, got 0.9999999999999999)
PASS softmax[2] (expected 0.6652409557748219, got 0.6652409557748218)
PASS softmax of huge scores (expected 0.5, got 0.5)
PASS tanh(0.5) (expected 0.46211715726000974, got 0.4621171572600098)
PASS tanh(100) (expected 1.0, got 1.0)
PASS tanh(-100) (expected -1.0, got -1.0)
PASS argmax
PASS fixed
PASS percent
PASS gauss mean (expected 0.0, got -0.002324214057947353)
PASS gauss variance (expected 1.0, got 0.9940411008047285)
PASS same seed, same numbers
PASS symbol_id
PASS symbol_at
PASS is_lowercase_word
PASS apple gives 6 examples
PASS first example
PASS last example ends the word
PASS 4083 parameters
PASS context_ids
PASS context_ids keeps the last letters
PASS untrained loss is near ln 27 (expected 3.295836866004329, got 3.2648090516679584)
PASS gradient of b2
PASS gradient of W2
PASS gradient of b1
PASS gradient of W1
PASS gradient of C
---
29 passed, 0 failed
```

All the gradients match. To convince yourself that the check really catches mistakes, break the backward pass on purpose. For example, in step C, change `(1.0 - hj * hj)` to `(1.0 - hj)`, and run the checks again: the gradients of W_1, $\mathbf{b}_1$ and C now fail, while those of W_2 and $\mathbf{b}_2$, which are computed before step C, still pass. Put it back before you go on.

Chapter 15

Taking a step

With the gradients in hand, the update is the easy part. For every parameter w, gradient descent does $w \leftarrow w - \eta\, g_w$, where g_w is the average gradient over the minibatch. The container holds the *sum* of the gradients of the batch, so the average is that sum divided by the batch size B:

$$w \;\leftarrow\; w - \frac{\eta}{B}\Big(\sum_{\text{examples}} \frac{\partial L}{\partial w}\Big).$$

That is `add_scaled` once more, with the factor $s = -\eta / B$. A matrix is updated one row at a time:

`next_letter_model.nex` – add at the end

```
-- m := m + s * g, row by row.
function update_matrix(m: Array[Array[Real]], s: Real, g: Array[Array[Real]])
  require
    same_rows: m.length = g.length
  do
    from
      let r: Integer := 0
    until
      r = m.length
    do
      add_scaled(m.get(r), s, g.get(r))
      r := r + 1
    end
  end
```

`next_letter_model.nex` – add before num_parameters

```
    -- Gradient descent: p := p - lr * (average gradient), for every
    -- parameter. g holds sums over the batch, so divide by its size.
    step(g: Gradients, lr: Real, batch_size: Integer)
      require
        positive_rate: lr > 0.0
        positive_batch: batch_size > 0
      do
        let s: Real := -lr / batch_size.to_real
        update_matrix(c, s, g.c)
        update_matrix(w1, s, g.w1)
        add_scaled(b1, s, g.b1)
        update_matrix(w2, s, g.w2)
        add_scaled(b2, s, g.b2)
      end
```

15.1 The first learning

We have every piece of training now: forward, loss, backward, step. Before we train on a whole dictionary, let's teach the model three words, and watch the loss fall. Each step uses all 18

examples of the three words as its batch.

Try it `try_learning.nex`

```
intern Next_Letter_Model
intern Training_Data
intern Helpers

let con: Console := create Console
let model: Next_Letter_Model := create Next_Letter_Model.make(4, 8, 64, 42)
let examples: Array[Example] := make_examples(["apple", "apply", "maple"], 4)
from
  let step: Integer := 0
until
  step > 300
do
  if step % 50 = 0 then
    con.print_line("step " + pad_left(step.to_string, 3) + ": loss "
      + fixed(evaluate(model, examples).loss, 4))
  end
  let g: Gradients := create Gradients.make(model)
  across examples as ex do
    model.backward(model.forward(ex.ids), ex.target, g)
  end
  model.step(g, 0.5, examples.length)
  step := step + 1
end
let probs: Array[Real] := model.forward(model.context_ids("appl")).probs
con.print_line("after \"appl\": e " + percent(probs.get(symbol_id("e")))
  + ", y " + percent(probs.get(symbol_id("y"))))
```

```
$ nex try_learning.nex

step   0: loss 3.2708
step  50: loss 0.1941
step 100: loss 0.1995
step 150: loss 0.1886
step 200: loss 0.1875
step 250: loss 0.1866
step 300: loss 0.1860
after "appl": e 50%, y 50%
```

It learns. The loss falls from about 3.3, knowing nothing, to a fraction of that. After `appl`, the model has learned that `e` (from `apple`) and `y` (from `apply`) are the two possibilities, about equally likely, which is exactly right for these three words. The loss cannot go to 0: with `appl` as the context, no model can know for certain which of the two words is meant.

Let's add this as a check:

`checks.nex` – add before the summary

```
-- Training lowers the loss: learn three words by heart.
let tiny: Array[Example] := make_examples(["apple", "apply", "maple"], 4)
let before: Real := evaluate(model, tiny).loss
repeat 300 do
  let g: Gradients := create Gradients.make(model)
```

```
  across tiny as t do
    model.backward(model.forward(t.ids), t.target, g)
  end
  model.step(g, 0.5, tiny.length)
end
let after: Real := evaluate(model, tiny).loss
c.check("training lowers the loss (" + fixed(before, 3) + " -> " + fixed(after, 3) + ")",
  after < before / 4.0)
```

Chapter 16

Saving and loading

Training on a dictionary takes a few minutes, and we don't want to do it every time we use the model. So we save the trained model to a file, and load it again in the program we play with.

We use JSON, a common text format for data, through Nex's `data/Json` library. A model becomes a JSON object with its three sizes and five parameter arrays:

```
{"context":4,"emb_dim":8,"hidden":64,"C":[[-0.38,...],...],"W1":[...],...}
```

`stringify` turns a Nex `Map` into JSON text. Its values can be anything, so the map's type is `Map[String, Any]`. Writing the file is `write_text` on a `Path`.

`next_letter_model.nex` – add before the invariant

```
    -- Write the model as JSON. The field names match the Python version
    -- of this project, so a model trained by one can be used by the other.
    save(path: String) do
      let data: Map[String, Any] := {
        "context": context, "emb_dim": emb_dim, "hidden": hidden,
        "C": c, "W1": w1, "b1": b1, "W2": w2, "b2": b2
      }
      let json: Json := create Json.make()
      create Path.make(path).write_text(json.stringify(data))
    end
```

Loading is the reverse, with one complication. `parse` can return any kind of JSON value, so its result has type `Any`, and so does everything inside it. To use a value as a number or an array, we must narrow it to that type with `convert`, which checks the actual type at run time: `if convert value to n: Integer then ... end` runs the `then` branch, with `n` as an `Integer`, only if the value really is one. Four small functions do this for each shape of value we expect, and raise an error for anything else:

`next_letter_model.nex` – add at the end

```
-- Reading numbers back out of parsed JSON, where every value is an Any.

function json_integer(value: Any): Integer do
  if convert value to n: Integer then
    result := n
  else
    raise "expected an integer in the model file"
  end
end

function json_real(value: Any): Real do
  if convert value to r: Real then
    result := r
  elseif convert value to n: Integer then
```

```
    result := n.to_real   -- a JSON number written without a decimal point
  else
    raise "expected a number in the model file"
  end
end

function json_vector(value: Any): Array[Real] do
  if convert value to items: Array[Any] then
    across items as item do
      result.add(json_real(item))
    end
  else
    raise "expected a list of numbers in the model file"
  end
end

function json_matrix(value: Any): Array[Array[Real]] do
  if convert value to rows: Array[Any] then
    across rows as row do
      result.add(json_vector(row))
    end
  else
    raise "expected a list of rows in the model file"
  end
end
```

Loading a model is a second way of creating one, so it is a second constructor, next to `make`: `create Next_Letter_Model.from_file("model.json")`. The class invariant is checked when it finishes, so a file with the wrong shapes is rejected right away.

`next_letter_model.nex` – add as a second constructor, after make

```
    -- A trained model, read back from a file written by save.
    from_file(path: String) do
      let json: Json := create Json.make()
      let text: String := create Path.make(path).read_text()
      if convert json.parse(text) to data: Map[String, Any] then
        context := json_integer(data.get("context"))
        emb_dim := json_integer(data.get("emb_dim"))
        hidden := json_integer(data.get("hidden"))
        c := json_matrix(data.get("C"))
        w1 := json_matrix(data.get("W1"))
        b1 := json_vector(data.get("b1"))
        w2 := json_matrix(data.get("W2"))
        b2 := json_vector(data.get("b2"))
      else
        raise "not a model file: " + path
      end
    end
```

One last detail before we can test this. The `nex` command starts programs from its own installation folder, so a file name like `model.json` would be looked for there, not in our project folder. `nex` records the folder it was started from in the environment variable `NEX_USER_DIR`, and we use that to turn a relative file name into a full one. A program run some other way has no `NEX_USER_DIR` (`getenv` gives the empty string), and then we leave the name as it is, relative to the current folder:

helpers.nex – add at the end

```
-- The full path of a file the user named. The `nex` launcher runs
-- programs from its own installation folder, so a relative name is
-- resolved against the folder `nex` was started from, NEX_USER_DIR.
-- Run some other way (no NEX_USER_DIR), a relative name is left as it is,
-- relative to the current folder.
function user_path(raw: String): String do
  let user_dir: String := (create Process).getenv("NEX_USER_DIR")
  if raw.starts_with("/") or user_dir = "" then
    result := raw
  else
    result := user_dir + "/" + raw
  end
end
```

The check saves the model we trained in the previous chapter, loads it back, compares the parameters, and deletes the file:

checks.nex – add before the summary

```
-- Saving and loading give back the same model.
let file: String := user_path("checks_model.json")
model.save(file)
let loaded: Next_Letter_Model := create Next_Letter_Model.from_file(file)
c.check("save and load", loaded.w1 = model.w1 and loaded.c = model.c
  and loaded.b2 = model.b2)
create Path.make(file).delete()
```

```
$ nex checks.nex

PASS dot (expected 32.0, got 32.0)
PASS affine
PASS softmax sums to 1 (expected 1.0, got 0.9999999999999999)
PASS softmax[2] (expected 0.6652409557748219, got 0.6652409557748218)
PASS softmax of huge scores (expected 0.5, got 0.5)
PASS tanh(0.5) (expected 0.46211715726000974, got 0.4621171572600098)
PASS tanh(100) (expected 1.0, got 1.0)
PASS tanh(-100) (expected -1.0, got -1.0)
PASS argmax
PASS fixed
PASS percent
PASS gauss mean (expected 0.0, got -0.002324214057947353)
PASS gauss variance (expected 1.0, got 0.9940411008047285)
PASS same seed, same numbers
PASS symbol_id
PASS symbol_at
PASS is_lowercase_word
PASS apple gives 6 examples
PASS first example
PASS last example ends the word
PASS 4083 parameters
PASS context_ids
PASS context_ids keeps the last letters
PASS untrained loss is near ln 27 (expected 3.295836866004329, got 3.2648090516679584)
PASS gradient of b2
```

```
PASS gradient of W2
PASS gradient of b1
PASS gradient of W1
PASS gradient of C
PASS training lowers the loss (3.271 -> 0.186)
PASS save and load
---
31 passed, 0 failed
```

Chapter 17

The training program

Time to put everything together into a program that trains a model on a whole word list.

17.1 Training and validation

There is a trap in measuring a model on the same examples it learned from: a model with enough parameters could memorize them, and score well without having learned anything general. That is called *overfitting*. So we hold back 10% of the words, never train on them, and measure the model on those, the *validation* set. If the training loss keeps falling while the validation loss starts rising, the model is memorizing.

We split *words*, not examples. If we split examples, the model could train on `.app` → `l` and be tested on `appl` → `e` from the same word, which is a little like seeing the exam questions in advance.

17.2 The settings

The program's settings, with their defaults, are read from the command line: `nex train.nex --steps 5000 --hidden 128`, say. Start the file with the interns, and a class that holds the settings:

New file `train.nex`

```
-- train.nex: train the next-letter model on a list of words.
--
--   nex train.nex                        -- uses /usr/share/dict/words
--   nex train.nex --words mywords.txt    -- any file with one word per line
--   nex train.nex --steps 5000           -- quicker, rougher model
--
-- It saves the trained model to model.json (change this with --out). The
-- model is also saved at every progress report, so stopping early with
-- Ctrl-C keeps the model as of the last report.
--
-- Each training step is the classic loop:
--   forward pass -> loss -> backpropagation -> parameter update
-- on a small random minibatch (stochastic gradient descent).

intern Vector_Math
intern Vocabulary
intern Rng
intern Next_Letter_Model
intern Training_Data
intern Helpers
intern io/Path
intern time/Date_Time

-- The settings of one training run, read from the command line.
```

```
class Options
  create
    from_command_line(args: Array[String]) do
      words := "/usr/share/dict/words"
      out := "model.json"
      steps := 20000
      batch := 32
      lr := 0.3
      context := 4
      emb := 8
      hidden := 64
      seed := 42
```

The constructor continues by going through the arguments in pairs, an option name and its value. `case … of` picks the branch whose string matches.

train.nex — continued

```
      from
        let i: Integer := 0
      until
        i >= args.length
      do
        if i + 1 >= args.length then
          raise "missing value after " + args.get(i)
        end
        let value: String := args.get(i + 1)
        case args.get(i) of
          "--words"   then words := value
          "--out"     then out := value
          "--steps"   then steps := value.to_integer
          "--batch"   then batch := value.to_integer
          "--lr"      then lr := value.to_real
          "--context" then context := value.to_integer
          "--emb"     then emb := value.to_integer
          "--hidden"  then hidden := value.to_integer
          "--seed"    then seed := value.to_integer
          else raise "unknown option " + args.get(i)
        end
        i := i + 2
      end
    end
  feature
    words: String    -- word list, one word per line
    out: String      -- where to save the trained model
    steps: Integer   -- number of minibatch updates
    batch: Integer   -- minibatch size
    lr: Real         -- learning rate (a tenth of it for the last 25%)
    context: Integer -- how many previous letters the model sees
    emb: Integer     -- embedding size
    hidden: Integer  -- hidden neurons
    seed: Integer    -- random seed, so runs can be repeated
  invariant
    positive_sizes: steps > 0 and batch > 0 and context > 0 and emb > 0 and hidden > 0
    positive_rate: lr > 0.0
end
```

The invariant rejects nonsense settings, such as `--steps 0` or a negative learning rate, before any work is done. We'll also want to show how long training has taken:

`train.nex` — add at the end

```
-- Seconds since `start`, a time in milliseconds.
function seconds_since(start: Integer): Real do
  result := (create Date_Time.now().epoch_millis() - start).to_real / 1000.0
end
```

17.3 Preparing the data

The main function starts by reading the settings and the word list, and splitting the words. `take(n)` gives the first n elements of an array, and `drop(n)` all the others.

`train.nex` — add at the end

```
function main() do
  let con: Console := create Console
  let opts: Options := create Options.from_command_line((create Process).command_line())
  let rng: Rng := create Rng.make(opts.seed)

  let words_path: Path := create Path.make(user_path(opts.words))
  if not words_path.exists() then
    con.print_line("Word list not found: " + opts.words)
    con.print_line("Pass one with --words (one word per line).")
    exit(1)
  end
  let words: Array[String] := load_words(words_path.path)
  if words.length < 10 then
    con.print_line("Too few usable lowercase words in " + opts.words)
    exit(1)
  end

  -- Hold back 10% of the WORDS for validation. Then we measure how well
  -- the model does on words it has never seen.
  shuffle(rng, words)
  let n_val: Integer := words.length / 10
  let val_words: Array[String] := words.take(n_val)
  let train_words: Array[String] := words.drop(n_val)
```

Then it makes the examples and the model, and reports the starting loss. Evaluating on all the validation examples after every thousand steps would slow training down, so we evaluate on a random sample of 3000, which is close enough.

`train.nex` — continued

```
  let train: Array[Example] := make_examples(train_words, opts.context)
  let val: Array[Example] := make_examples(val_words, opts.context)
  -- Evaluating on 3000 random validation examples is quick and close enough.
  shuffle(rng, val)
  let val_sample: Array[Example] := val.take(3000)

  let model: Next_Letter_Model :=
```

```
    create Next_Letter_Model.make(opts.context, opts.emb, opts.hidden, opts.seed)
  con.print_line(train_words.length.to_string + " training words -> "
    + train.length.to_string + " examples; "
    + val_words.length.to_string + " validation words")
  con.print_line("model has " + model.num_parameters.to_string + " parameters (context="
    + opts.context.to_string + ", emb=" + opts.emb.to_string
    + ", hidden=" + opts.hidden.to_string + ")")

  let before: Evaluation := evaluate(model, val_sample)
  con.print_line("before training: val loss " + fixed(before.loss, 3)
    + " (guessing uniformly would give " + fixed(Vocabulary.SIZE.to_real.log, 3) + ")")
  con.print_line("")
```

17.4 The training loop

Each step builds a minibatch of randomly chosen training examples, runs forward and backward on each, and updates the model once with their summed gradients. These few lines are the heart of the whole project.

We use one more trick, a *learning-rate schedule*: the full learning rate for the first 75% of the steps, then a tenth of it. Big steps make fast progress early on, but near the bottom of the valley they bounce from side to side, like the $\eta = 1.1$ example of Chapter 12 but milder. Smaller steps let the model settle.

train.nex — continued

```
  let report_every: Integer := (opts.steps / 20).max(1)
  let running_loss: Real := 0.0
  let start: Integer := create Date_Time.now().epoch_millis()

  from
    let step: Integer := 1
  until
    step > opts.steps
  do
    -- Learning-rate schedule: big steps first for fast progress, then
    -- smaller steps to settle into the bottom of the valley.
    let lr: Real := when step * 4 <= opts.steps * 3 then opts.lr else opts.lr / 10.0 end

    -- One minibatch of randomly chosen examples: the "stochastic" in
    -- stochastic gradient descent.
    let grads: Gradients := create Gradients.make(model)
    repeat opts.batch do
      let ex: Example := train.get(rng.below(train.length))
      let fp: Forward_Pass := model.forward(ex.ids)                      -- forward
      running_loss := running_loss + cross_entropy(fp.probs, ex.target) -- loss
      model.backward(fp, ex.target, grads)                               -- backprop
    end
    model.step(grads, lr, opts.batch)                                    -- update
```

Twenty times during training, the loop prints a progress report: the average loss on the training examples since the last report, the loss and accuracy on the validation sample, and the time so far. It also saves the model, so that stopping early with Ctrl-C loses at most one report's worth of training.

train.nex — continued

```
    if step % report_every = 0 or step = opts.steps then
      let train_loss: Real := running_loss / (report_every * opts.batch).to_real
      running_loss := 0.0
      let val_now: Evaluation := evaluate(model, val_sample)
      con.print_line("step " + pad_left(step.to_string, 6) + "/" + opts.steps.to_string
        + "  lr " + fixed(lr, 3)
        + "  train loss " + fixed(train_loss, 3)
        + "  val loss " + fixed(val_now.loss, 3)
        + "  val accuracy " + pad_left(fixed(val_now.accuracy * 100.0, 1), 4) + "%"
        + "  [" + pad_left(fixed(seconds_since(start), 0), 5) + "s]")
      model.save(user_path(opts.out))
    end
    step := step + 1
  end

  con.print_line("")
  con.print_line("saved to " + opts.out + ". Try:  nex play.nex")
end
```

And finally, the line that runs it all:

train.nex — add at the end

```
main()
```

17.5 Running it

A quick run first, with a tenth of the default steps, to check that everything works:

```
$ nex train.nex --steps 2000 --out quick.json

57465 training words -> 533799 examples; 6384 validation words
model has 4083 parameters (context=4, emb=8, hidden=64)
before training: val loss 3.297 (guessing uniformly would give 3.296)

step    100/2000  lr 0.300  train loss 2.872  val loss 2.717  val accuracy 20.6%  [    1s]
step    200/2000  lr 0.300  train loss 2.637  val loss 2.585  val accuracy 22.4%  [    2s]
step    300/2000  lr 0.300  train loss 2.559  val loss 2.563  val accuracy 22.9%  [    3s]
step    400/2000  lr 0.300  train loss 2.532  val loss 2.512  val accuracy 24.3%  [    4s]
step    500/2000  lr 0.300  train loss 2.488  val loss 2.516  val accuracy 24.9%  [    5s]
step    600/2000  lr 0.300  train loss 2.454  val loss 2.468  val accuracy 25.8%  [    6s]
step    700/2000  lr 0.300  train loss 2.457  val loss 2.438  val accuracy 26.2%  [    7s]
step    800/2000  lr 0.300  train loss 2.418  val loss 2.455  val accuracy 25.8%  [    8s]
step    900/2000  lr 0.300  train loss 2.416  val loss 2.454  val accuracy 25.6%  [    9s]
step   1000/2000  lr 0.300  train loss 2.431  val loss 2.392  val accuracy 27.5%  [    9s]
step   1100/2000  lr 0.300  train loss 2.415  val loss 2.386  val accuracy 27.5%  [   10s]
step   1200/2000  lr 0.300  train loss 2.392  val loss 2.415  val accuracy 26.2%  [   11s]
step   1300/2000  lr 0.300  train loss 2.384  val loss 2.384  val accuracy 27.7%  [   12s]
step   1400/2000  lr 0.300  train loss 2.378  val loss 2.388  val accuracy 27.2%  [   13s]
step   1500/2000  lr 0.300  train loss 2.382  val loss 2.359  val accuracy 27.4%  [   14s]
step   1600/2000  lr 0.030  train loss 2.322  val loss 2.318  val accuracy 27.9%  [   15s]
step   1700/2000  lr 0.030  train loss 2.291  val loss 2.304  val accuracy 28.8%  [   16s]
step   1800/2000  lr 0.030  train loss 2.273  val loss 2.294  val accuracy 29.0%  [   16s]
```

```
step   1900/2000  lr 0.030  train loss 2.301  val loss 2.291  val accuracy 29.5%  [   17s]
step   2000/2000  lr 0.030  train loss 2.295  val loss 2.287  val accuracy 29.8%  [   18s]

saved to quick.json. Try:  nex play.nex
```

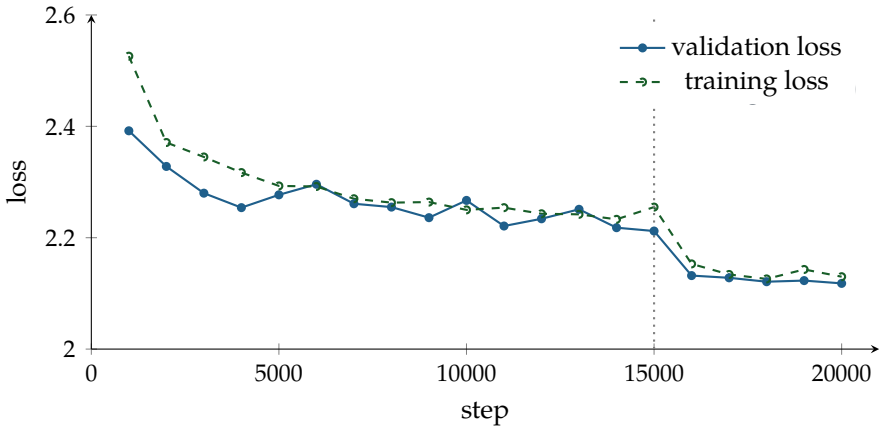
Now the real thing. With the defaults, 20000 steps of 32 examples, the model sees 640,000 examples, a little more than the whole training set once. It takes a few minutes:

```
nex train.nex
```

```
57465 training words -> 533799 examples; 6384 validation words
model has 4083 parameters (context=4, emb=8, hidden=64)
before training: val loss 3.297 (guessing uniformly would give 3.296)

step   1000/20000  lr 0.300  train loss 2.526  val loss 2.392  val accuracy 27.5%  [   10s]
step   2000/20000  lr 0.300  train loss 2.371  val loss 2.328  val accuracy 28.8%  [   17s]
step   3000/20000  lr 0.300  train loss 2.345  val loss 2.280  val accuracy 29.0%  [   25s]
...
step  14000/20000  lr 0.300  train loss 2.233  val loss 2.218  val accuracy 29.8%  [  118s]
step  15000/20000  lr 0.300  train loss 2.255  val loss 2.212  val accuracy 31.2%  [  127s]
step  16000/20000  lr 0.030  train loss 2.153  val loss 2.132  val accuracy 32.3%  [  138s]
step  17000/20000  lr 0.030  train loss 2.134  val loss 2.128  val accuracy 32.7%  [  148s]
step  18000/20000  lr 0.030  train loss 2.126  val loss 2.121  val accuracy 32.9%  [  157s]
step  19000/20000  lr 0.030  train loss 2.143  val loss 2.123  val accuracy 32.9%  [  166s]
step  20000/20000  lr 0.030  train loss 2.130  val loss 2.118  val accuracy 33.1%  [  174s]

saved to model.json. Try:  nex play.nex
```



How to read this:

- **The loss falls fast, then slowly.** From 3.30 (knowing nothing) to 2.39 within the first thousand steps, and then only slowly to 2.21 by step 15000, bouncing up and down because every minibatch is different.
- **The learning-rate drop helps at once.** At step 16000, just after the drop, the loss falls from about 2.21 to 2.13: the smaller steps settle into the valley instead of bouncing around in it.

- **No overfitting.** The training and validation losses stay close together all the way. With 4083 parameters and over half a million examples, the model cannot memorize the words; it can only learn general spelling patterns. In fact a model this small is more likely *underfitting*: too small to capture everything there is to learn. Try `--hidden 128`.
- **What the numbers mean.** The final loss of 2.12 is a perplexity of $e^{2.12} \approx 8.3$: on average, the model is about as unsure as if it were choosing among 8 equally likely symbols, instead of 27. Its top guess is right 33% of the time, against 3.7% for guessing. That might sound low, but even a perfect model would be far from 100%: after `app`, both `l` (apple) and `r` (approve) are right for some words.

Chapter 18

Playing with the model

The last program loads the trained model and uses it to complete words.

18.1 Predictions

First, a method that gives every symbol with its probability, most likely first. A small class holds one such pair:

next_letter_model.nex — add before the model class

```
-- One row of a prediction: a symbol and its probability.
class Prediction
  create
    make(symbol: String, probability: Real) do
      this.symbol := symbol
      this.probability := probability
    end
  feature
    symbol: String
    probability: Real
end
```

`sort` takes a comparison function, written here with `fn`: it returns a negative number if `a` should come first, and a positive one if `b` should. Comparing `b` with `a`, rather than `a` with `b`, sorts from most to least likely.

next_letter_model.nex — add before num_parameters

```
    -- Every symbol with its probability, most likely first.
    predict(prefix: String): Array[Prediction]
      require
        known_symbols: is_lowercase_word(prefix) or prefix = ""
      do
        let probs: Array[Real] := forward(context_ids(prefix)).probs
        let unsorted: Array[Prediction] := []
        from
          let i: Integer := 0
        until
          i = Vocabulary.SIZE
        do
          unsorted.add(create Prediction.make(symbol_at(i), probs.get(i)))
          i := i + 1
        end
        result := unsorted.sort(fn(a, b: Prediction): Integer do
          result := b.probability.compare(a.probability)
        end)
      ensure
        every_symbol: result.length = Vocabulary.SIZE
      end
```

One more check, that the model trained on three words in Chapter 15 has learned the possible continuations of `appl`:

`checks.nex` — add before the summary

```
-- Predictions come sorted, and say what was learned.
let ranked: Array[Prediction] := model.predict("appl")
c.check("predictions are sorted", ranked.get(0).probability >= ranked.get(1).probability
  and ranked.get(1).probability >= ranked.get(26).probability)
c.check("it learned: after \"appl\" comes e or y",
  ranked.get(0).symbol = "e" or ranked.get(0).symbol = "y")
```

```
$ nex checks.nex

PASS dot (expected 32.0, got 32.0)
PASS affine
PASS softmax sums to 1 (expected 1.0, got 0.9999999999999999)
PASS softmax[2] (expected 0.6652409557748219, got 0.6652409557748218)
PASS softmax of huge scores (expected 0.5, got 0.5)
PASS tanh(0.5) (expected 0.46211715726000974, got 0.4621171572600098)
PASS tanh(100) (expected 1.0, got 1.0)
PASS tanh(-100) (expected -1.0, got -1.0)
PASS argmax
PASS fixed
PASS percent
PASS gauss mean (expected 0.0, got -0.002324214057947353)
PASS gauss variance (expected 1.0, got 0.9940411008047285)
PASS same seed, same numbers
PASS symbol_id
PASS symbol_at
PASS is_lowercase_word
PASS apple gives 6 examples
PASS first example
PASS last example ends the word
PASS 4083 parameters
PASS context_ids
PASS context_ids keeps the last letters
PASS untrained loss is near ln 27 (expected 3.295836866004329, got 3.2648090516679584)
PASS gradient of b2
PASS gradient of W2
PASS gradient of b1
PASS gradient of W1
PASS gradient of C
PASS training lowers the loss (3.271 -> 0.186)
PASS save and load
PASS predictions are sorted
PASS it learned: after "appl" comes e or y
---
33 passed, 0 failed
```

That is the complete set of checks.

18.2 The play program

The program has two modes. Given some word beginnings on the command line, it completes each one by always taking the model's top choice. With no arguments, it lets you steer, as in the session at the start of this tutorial. The file starts with the interns and two constants:

New file `play.nex`

```
-- play.nex: type the start of a word and let the model finish it, letter
-- by letter.
--
--   nex play.nex                    -- interactive
--   nex play.nex app hel qu         -- complete each prefix and exit
--   nex play.nex --model other.json
--
-- Interactive keys, after each suggestion:
--   Enter      accept the suggested letter
--   n          show the next most likely letter instead
--   letters    type them yourself (e.g. "r" to steer "app" toward "appr...")
--   q          give up on this word and start a new one

intern Vocabulary
intern Next_Letter_Model
intern Helpers
intern io/Path

class Play
  feature
    MAX_LEN = 30        -- safety stop, in case the model never predicts "."
    ALTERNATIVES = 5    -- how many suggestions "n" cycles through
end
```

Completing a word by always taking the top choice is called *greedy* decoding. It stops when the top choice is the end marker, or, as a safety net, when the word gets very long.

`play.nex` – add at the end

```
function describe(symbol: String): String do
  result := when symbol = Vocabulary.END then "<end of word>" else symbol end
end

-- Always take the single most likely symbol until the model says ".".
function greedy_complete(model: Next_Letter_Model, prefix: String): String do
  result := prefix
  let done: Boolean := false
  from
  until
    done or result.length >= Play.MAX_LEN
  do
    let best: String := model.predict(result).get(0).symbol
    if best = Vocabulary.END then
      done := true
    else
      result := result + best
    end
  end
end
```

For the interactive mode, two helpers: one lists the other top choices, and one reads a line of input. `read_line` returns `nil` at the end of the input (when you press Ctrl-D), so its type is `?String`, "a string or nil". The *object test* `?line as text` is true when `line` is not `nil`, and then gives us the string as `text`.

`play.nex` — add at the end

```
-- The other top choices, e.g. "r 21%, e 5%, end 3%".
function others(ranked: Array[Prediction], skip: String): String do
  across ranked.take(Play.ALTERNATIVES) as p do
    if p.symbol /= skip then
      if result /= "" then
        result := result + ", "
      end
      result := result + when p.symbol = Vocabulary.END then "end" else p.symbol end
        + " " + percent(p.probability)
    end
  end
end

-- Read a line, lowercased and trimmed; nil at end of input (Ctrl-D).
function ask(con: Console, prompt: String): ?String do
  let line: ?String := con.read_line(prompt)
  if ?line as text then
    result := text.trim().to_lower()
  end
end
```

The interactive completion of one word shows a suggestion, reads your answer, and acts on it, until the word is finished, abandoned, or the input ends. `choice` says which of the ranked options is being shown; `n` moves it to the next one.

`play.nex` — add at the end

```
-- Suggest letters for one word until it is finished or abandoned.
-- Returns false when the input has ended.
function complete_interactively(con: Console, model: Next_Letter_Model,
                                start: String): Boolean do
  result := true
  let word: String := start
  let choice: Integer := 0      -- which of the ranked options to show
  let finished: Boolean := false
  let abandoned: Boolean := false
  from
  until
    finished or abandoned or not result or word.length >= Play.MAX_LEN
  do
    let ranked: Array[Prediction] := model.predict(word)
    let shown: Prediction := ranked.get(choice)
    let prompt: String := when shown.symbol = Vocabulary.END then
      "  " + word + "   -> looks complete (" + percent(shown.probability) + ")   "
    else
      "  " + word + "[" + shown.symbol + "]   (" + percent(shown.probability) + ")   "
    end
    con.print_line(prompt + "   others: " + others(ranked, shown.symbol))
```

play.nex – continued

```
    let answer: ?String := ask(con, "  > ")
    if ?answer as ans then
      if ans = "" then                          -- accept the suggestion
        if shown.symbol = Vocabulary.END then
          finished := true
        else
          word := word + shown.symbol
          choice := 0
        end
      elseif ans = "n" then                     -- show the next option
        choice := (choice + 1) % Play.ALTERNATIVES
      elseif ans = "q" then                     -- abandon this word
        abandoned := true
      elseif is_lowercase_word(ans) then        -- the user types letters
        word := word + ans
        choice := 0
      else
        con.print_line("  Enter = accept, n = next option, letters = type your own, "
          + "q = new word")
      end
    else
      result := false                           -- end of input
    end
  end
  if abandoned or not result then
    con.print_line("")
  else
    con.print_line("  ==> " + word)
    con.print_line("")
  end
end
```

The outer loop asks for the start of a word, and hands it to `complete_interactively`, until you enter an empty line or the input ends:

play.nex – add at the end

```
function interactive(con: Console, model: Next_Letter_Model) do
  con.print_line("Next-letter predictor. Type the start of a word "
    + "(empty line or Ctrl-D to quit).")
  con.print_line("After each suggestion: Enter = accept, n = next option, "
    + "letters = type your own, q = new word.")
  con.print_line("")
  let going: Boolean := true
  from
  until
    not going
  do
    let line: ?String := ask(con, "start of word> ")
    if ?line as word then
      if word = "" then
        going := false
      elseif not is_lowercase_word(word) then
        con.print_line("  please use letters a-z only")
        con.print_line("")
```

```
      else
        going := complete_interactively(con, model, word)
      end
    else
      con.print_line("")
      going := false
    end
  end
end
```

Finally, `main` reads the command line (an optional `--model` file, and any word beginnings), loads the model, and runs one of the two modes.

play.nex — add at the end

```
function main() do
  let con: Console := create Console
  let args: Array[String] := (create Process).command_line()
  let model_file: String := "model.json"
  let prefixes: Array[String] := []
  from
    let i: Integer := 0
  until
    i >= args.length
  do
    if args.get(i) = "--model" and i + 1 < args.length then
      model_file := args.get(i + 1)
      i := i + 2
    else
      prefixes.add(args.get(i).to_lower())
      i := i + 1
    end
  end

  if not create Path.make(user_path(model_file)).exists() then
    con.print_line("No model at " + model_file + ". Train one first: nex train.nex")
    exit(1)
  end
  let model: Next_Letter_Model :=
    create Next_Letter_Model.from_file(user_path(model_file))
```

play.nex — continued

```
  if prefixes.is_empty then
    interactive(con, model)
  else
    across prefixes as prefix do
      if is_lowercase_word(prefix) then
        let ranked: Array[Prediction] := model.predict(prefix)
        let top: String := ""
        across ranked.take(3) as p do
          if top /= "" then
            top := top + ", "
          end
          top := top + describe(p.symbol) + " " + percent(p.probability)
        end
```

```
        con.print_line(pad_left(prefix, 12) + " -> "
          + pad_right(greedy_complete(model, prefix), 16) + "(next letter: " + top + ")")
      else
        con.print_line(pad_left(prefix, 12) + " -> please use letters a-z only")
      end
    end
  end
end

main()
```

18.3 Trying it

With the model trained in the previous chapter:

```
$ nex play.nex app thin walk qu

         app -> apprate         (next letter: r 23%, e 19%, l 17%)
        thin -> thing           (next letter: g 72%, e 5%, d 5%)
        walk -> walking         (next letter: i 26%, e 24%, s 19%)
          qu -> quint           (next letter: i 27%, a 23%, e 19%)
```

After `thin`, the model is 72% sure of `g`, and then sure that `thing` is complete. After `walk`, it suggests `walking`. But `app` became `apprate` and `qu` became `quint`: always taking the top choice tends to produce plausible-looking words that are not quite real, or are not the ones you meant. The interactive mode, where you steer, works much better:

```
nex play.nex
```

```
start of word> app
  app[r]   (23%)      others: e 19%, l 17%, o 15%, i 9%
  > l
  appl[e]   (45%)      others: i 19%, a 13%, y 11%, o 8%
  >
  apple[a]   (21%)      others: s 18%, r 11%, end 7%, n 7%
  > n
  apple[s]   (18%)      others: a 21%, r 11%, end 7%, n 7%
  >
  apples   -> looks complete (42%)      others: s 19%, t 15%, e 4%, h 4%
  >
  ==> apples
```

Why does the model prefer `r` after `app`, when `apple` is surely the most common word starting that way? Because it learned from a dictionary, which lists every word once. "Apple" and "appraise" count the same, and the dictionary has many more words starting with `appr` (approach, approve, appropriate, ...) and `appe` (appear, append, appetite, ...) than with `appl`. The model has learned the dictionary's spelling patterns, not how often people use each word. Trained on ordinary text, with common words appearing many times, it would learn different preferences.

Chapter 19

Where to go next

You have built, from nothing, every part of a neural network that learns to spell: embeddings, layers, an activation function, softmax, the cross-entropy loss, backpropagation, a gradient check, and stochastic gradient descent with a learning-rate schedule.

19.1 What it can't do

- **It only sees four letters.** In a long word, it has forgotten how the word began. Raise `--context` to see more, at the cost of more parameters and slower training.
- **It knows spelling, not usage**, as we saw with `app`.
- **It invents words.** It learned what English words look like, not a list of them, so it happily completes `qu` to something word-like that isn't in the dictionary. This is the same behavior that makes large language models "hallucinate": they produce what is plausible, which is not always what is true.

19.2 Experiments to try

- **Bigger models.** Train with `--hidden 128`, or `--emb 10 --context 5`, and compare the validation loss. Does the gap between training and validation loss grow?
- **Learning rates.** Try `--lr 0.1` and `--lr 1.0`. Which trains faster? Does either become unstable?
- **ReLU.** Replace tanh with the *rectified linear unit*, $\mathrm{relu}(v) = \max(0, v)$, which is popular in deep networks. Its derivative is 1 for $v > 0$ and 0 otherwise. You need to change `forward` (the activation) and step C of `backward` (the derivative); then run the gradient check. (Strictly, the derivative at exactly $v = 0$ is undefined; any choice works in practice.)
- **Temperature.** Divide the scores by a number T before softmax: $p_i = e^{z_i/T} / \sum_j e^{z_j/T}$. With $T < 1$, the differences between scores grow and the model becomes more decisive; with $T > 1$, the probabilities even out and it becomes more adventurous. Then, instead of always taking the top choice, pick a symbol at random according to its probability (*sampling*), and let the model invent new words.
- **A second hidden layer.** Add a layer $\mathbf{h}_2 = \tanh(W_3\mathbf{h} + \mathbf{b}_3)$ between the hidden layer and the output. The backward pass gains one more step B and C. The gradient check will tell you whether you got it right.

19.3 From here to a language model

The models behind today's chatbots are built from the same pieces. Their *vocabulary* has tens of thousands of *tokens* (words and pieces of words) instead of 27 symbols, each with an embedding

of thousands of numbers. Instead of joining a fixed window of four embeddings, they use a mechanism called *attention* that lets every position look back at every earlier one, across thousands of tokens. They stack dozens of layers, and have billions of parameters, trained on a large part of the text on the internet with many computers working for months. But at the end there is still a softmax over the vocabulary, trained with cross-entropy to predict what comes next, with gradients computed by backpropagation and applied by a variant of gradient descent. You have written all of those by hand.

Appendix A

The math on one page

Forward pass, for context symbols $s_1, \dots, s_4$ and target t

$$\mathbf{x} = (C_{s_1}, C_{s_2}, C_{s_3}, C_{s_4}) \qquad \mathbf{a} = W_1\mathbf{x} + \mathbf{b}_1 \qquad \mathbf{h} = \tanh(\mathbf{a})$$

$$\mathbf{z} = W_2\mathbf{h} + \mathbf{b}_2 \qquad p_i = \frac{e^{z_i - m}}{\sum_j e^{z_j - m}},\ m = \max_j z_j \qquad L = -\ln p_t$$

Backward pass

$$\frac{\partial L}{\partial z_i} = p_i - [i = t]$$

$$\frac{\partial L}{\partial W_{2,ij}} = \frac{\partial L}{\partial z_i} h_j, \quad \frac{\partial L}{\partial b_{2,i}} = \frac{\partial L}{\partial z_i}, \quad \frac{\partial L}{\partial h_j} = \sum_i \frac{\partial L}{\partial z_i} W_{2,ij}$$

$$\frac{\partial L}{\partial a_j} = \frac{\partial L}{\partial h_j}(1 - h_j^2)$$

$$\frac{\partial L}{\partial W_{1,ij}} = \frac{\partial L}{\partial a_i} x_j, \quad \frac{\partial L}{\partial b_{1,i}} = \frac{\partial L}{\partial a_i}, \quad \frac{\partial L}{\partial x_j} = \sum_i \frac{\partial L}{\partial a_i} W_{1,ij}$$

$$\frac{\partial L}{\partial C_{s,m}} = \sum_{k\,:\,s_k = s} \frac{\partial L}{\partial x_{k\cdot d + m}} \qquad (d = \text{embedding size})$$

$[i = t]$ is 1 if $i = t$ and 0 otherwise.

Training

$$w \leftarrow w - \frac{\eta}{B} \sum_{\text{batch}} \frac{\partial L}{\partial w} \qquad \text{minibatch gradient descent, batch size } B$$

$$f'(w) \approx \frac{f(w+\varepsilon) - f(w-\varepsilon)}{2\varepsilon} \qquad \text{gradient check, error } O(\varepsilon^2)$$

Useful facts

$$\tanh(v) = 1 - \frac{2}{e^{2v} + 1}, \quad \tanh'(v) = 1 - \tanh^2(v)$$

$$\operatorname{softmax}(\mathbf{z} + c) = \operatorname{softmax}(\mathbf{z})$$

$$\operatorname{Var}\Big(\sum_{j=1}^{n} w_j x_j\Big) = n\,\sigma_w^2\,\sigma_x^2 \quad \text{(independent, mean 0), so use } \sigma_w = 1/\sqrt{n}$$

$$\text{know-nothing loss} = \ln 27 \approx 3.296, \quad \text{perplexity} = e^L$$

$$\text{Box–Muller:} \quad z = \sqrt{-2\ln u_1}\,\cos(2\pi u_2)$$

$$\text{Park–Miller:} \quad \text{state} \leftarrow 48271 \cdot \text{state} \bmod (2^{31} - 1)$$

Appendix B

The Nex you met

A reminder of the Nex features this project uses.

Feature	Example
`intern` another file	`intern Helpers` loads everything in `helpers.nex`; no class of that name is needed
Class constants	`SIZE = 27`, read as `Vocabulary.SIZE`
`result` and its zero value	`result` starts at `0.0`, `""`, `[]`, …
Preconditions, postconditions	`require same_length: ..., ensure sums_to_one: ...`
Class invariants	`invariant state_in_range: ...`
`rescue` and `exception`	catching a contract violation in a try-it script
`when ... then ... else ... end`	an `if` that gives a value
Loops	`from ... until ... do ... end`, `repeat n do`, `across xs as x do`
Generic functions	`shuffle[T](rng: Rng, items: Array[T])`
Several constructors	`make` and `from_file`
Anonymous functions	`fn(a, b: Prediction): Integer do ... end` for `sort`
`convert`	narrowing an `Any` from JSON to `Integer`, `Real`, `Array[Any]`
Detachable types, object test	`?String`, `if ?line as text then`
`case ... of`	matching option names in `train.nex`
Libraries	`data/Json`, `io/Path`, `time/Date_Time`, `Console`, `Process`